Transmission Capacity and Sequential Bottlenecks in Fiscal–Monetary Policy

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Abstract

Why did the largest peacetime fiscal-monetary stimulus in U.S. history—$9 trillion deployed in lockstep during 2020–21—deliver so little? We argue that the answer lies in the economy's *transmission capacity*— an endogenous state variable, not a fixed multiplier, that determines how much of a policy impulse survives the journey from injection to private expenditure. Using U.S. quarterly data (1991Q2–2024Q4, $T = 135$), we model transmission capacity as the product of three multiplicative channels—debt absorption, banking health, and a confidence multiplier—that act as sequential bottlenecks: liquidity must survive each stage before the next one can amplify it. When any single channel is constrained, capacity is limited regardless of the size of the impulse or the degree of coordination behind it. That is the mechanism. Here is the evidence.

We show that transmission exhibits non-additive collapses inconsistent with standard models. When all three channels are simultaneously impaired, transmission falls 55 percent below what an additive framework predicts. Three model-free tests reject pure additivity and support sequential bottleneck structure. Counterfactual analysis—read as illustrative of the mechanism rather than precise causal measurement—finds that restoring pre-pandemic *fiscal space* alone would raise COVID-era transmission by 82 percent; restoring pre-crisis *banking health* adds 9 percent; and perfect *coordination* adds 12 percent. An interaction-adjusted Shapley decomposition attributes approximately 70 percent of the pre-crisis-to-pandemic transmission loss to debt absorption, a ranking robust across 34 of 36 calibration cells. Local projections confirm that transmission capacity positively and persistently predicts output at horizons of one to four quarters. The binding constraint on transmission is structural: any model consistent with the data must feature strong channel interaction effects that purely additive specifications cannot generate.

JEL Codes: E52, E62, E63, H63.

Keywords: policy transmission; fiscal-monetary coordination; debt absorption; banking health; confidence multiplier; bottleneck; stabilisation capacity.

> "The recovery will be stronger and move faster if monetary policy and fiscal policy continue to work side by side to provide support to the economy until it is clearly out of the woods."
>
> *Jerome Powell, October 6, 2020*

1 Introduction

Consider what happened in 2020. The Federal Reserve cut rates to zero and launched $4.5 trillion in asset purchases. Congress passed $5 trillion in fiscal support—the CARES Act, the American Rescue Plan, and related programmes. The two authorities moved in near-perfect lockstep: the largest synchronised stimulus in modern peacetime history, equivalent to 25 percent of GDP. And yet GDP took five full quarters to recover to its pre-pandemic level. Employment remained 4 million jobs below trend through 2022. Investment plateaued far below its pre-crisis path despite record-low borrowing costs. More coordination than ever. Less transmission than expected.

This is not a COVID-19 anomaly. After the Global Financial Crisis, four years of zero interest rates and unprecedented asset purchases produced the "missing inflation" episode (Coibion and Gorodnichenko, 2015; Mavroeidis et al., 2014) and a recovery that economists described, with some embarrassment, as the slowest since the Great Depression. During the 2011–17 consolidation, transmission remained depressed even when the Fed provided generous accommodation. During the 2022–23 tightening cycle, the most aggressive contraction in four decades produced unusually slow disinflation. The pattern across these episodes is consistent: increasingly forceful interventions generate attenuated responses. The scale and synchronisation of these interventions rules out misalignment as the primary explanation. We acknowledge that slow recoveries also reflect supply disruptions, sectoral recomposition, and pandemic-specific shocks. But those forces vary across episodes, whereas the at-

tenuation of large coordinated interventions does not. The pattern is too consistent across the GFC, the austerity period, and 2020–21 to be episode-specific. Something more structural is at work.

The deeper question, then, is not about coordination. It is about *transmission capacity*: the fraction of a policy impulse that survives the journey from injection to private expenditure. Transmission capacity is not a fixed multiplier. It is an endogenous state variable that depends on the structural conditions of the economy at the moment the impulse arrives. We model it as the product of three sequential frictions that each attenuate the impulse before passing it to the next stage.

The first is *debt absorption*: at current debt levels, a large fraction of available liquidity is diverted to debt service before it touches private spending (Chudik et al., 2017; Blanchard, 2019). The second is *banking health*: what survives debt absorption must still be intermediated by banks whose balance sheets may be too impaired to translate liquidity into credit (Gertler and Karadi, 2011; Brunnermeier and Koby, 2024). The third is a *confidence multiplier*: private agents who doubt that policy will stabilise future income save rather than spend the credit they receive (Carroll, 2001; Evans and Honkapohja, 2001; Angeletos and Huo, 2021). These three frictions act as sequential bottlenecks. In an additive world, weakness in one channel partially offsets strength in another. In a multiplicative world—which the data are consistent with—weakness in any single channel is enough to limit transmission regardless of how forceful the policy impulse or how well-coordinated its delivery.

That distinction has a testable empirical implication (Corollary 1.1), and we test it three ways, each independent of the functional form we assume. When all three channels are simultaneously impaired, actual transmission is 55 percent below what an additive model would predict—a collapse that additivity structurally cannot generate. Interaction terms between the three channels account for 5.7 percent of the variance in estimated transmission effectiveness; any purely additive specification generates exactly zero interaction variance by construction. And the effect of banking health on transmission is 51 percentage points larger when fiscal space is ample

than when it is absorbed: the cross-partial gap is the fingerprint of multiplicative aggregation, since the additive model predicts a gap of zero. All three findings are consistent with the sequential bottleneck structure; they do not uniquely identify the multiplicative form—other nonlinearities such as regime shifts would also generate positive cross-partials— but they establish that channel interactions are empirically material. Any model consistent with the joint behaviour of transmission, debt, and financial conditions must feature strong interaction effects across channels: the additive null is empirically untenable in this context.

What we find. We estimate the model on 135 U.S. quarterly observations (1991Q2–2024Q4) using a two-stage GMM–Extended Kalman Filter procedure with seven external instruments. Three findings stand out.

First, transmission exhibits non-additive collapses inconsistent with standard models. Three model-free tests reject pure additivity and are consistent with sequential bottleneck structure: interaction terms account for 5.7 percent of transmission variance (zero in any additive model), simultaneous three-channel stress produces a 55 percent excess collapse beyond the additive prediction, and cross-partial correlations are 51 percentage points larger when complementary channels are healthy than when they are impaired.

Second, counterfactual analysis—which should be read as illustrative of the mechanism rather than precise causal measurement—shows that structural frictions dominate. Restoring pre-pandemic *fiscal space* would have raised COVID-era transmission by 82 percent under the model; restoring pre-crisis *banking health* adds 9 percent; *perfect coordination* adds 12 percent. The ranking is robust across 34 of 36 calibration cells, but the magnitudes depend on modelling choices and should not be taken as structural parameters.

Third, coordination is not neutral with respect to confidence. In normal times, institutional alignment reinforces private confidence: the estimated loading is $\hat{\alpha}_1 = 0.98$ ($t = 8.96$). During systemic crises, the regime shifts: $\hat{\alpha}_{1c} = -0.46$ ($t = -2.26$,

$p = 0.026$), reducing the effective loading to 0.52. When both fiscal and monetary authorities respond to an emergency in lockstep, private agents read coordination as a signal of the severity of the shock rather than as reassurance (Bianchi and Melosi, 2019; Bianchi et al., 2023). The policy implication is direct: even the confidence dividend of coordination shrinks precisely when it is most needed.

Related literature. The paper speaks to three literatures, and to a gap shared by all three: standard frameworks assume either fixed transmission multipliers or additive channel contributions, neither of which generates the non-additive collapses we document. On fiscal-monetary coordination, Leeper (1991), Sims (1994), and Bianchi and Melosi (2019) establish that regime alignment shapes multipliers; Bianchi et al. (2023) and Bredemeier et al. (2024) document how debt and financing conditions modulate these effects. We extend this strand by showing that when channels interact multiplicatively, the marginal value of improving any single channel depends on the state of the others— a prediction that standard additive frameworks rule out by construction. On banking intermediation, Gertler and Karadi (2011) and Brunnermeier and Koby (2024) establish the pass-through and reversal-rate mechanisms that motivate our χ_t state. On debt dynamics, Chudik et al. (2017) and Blanchard (2019) provide the nonlinear absorption evidence our δ_t function is calibrated to match. Our contribution is to embed all three frictions in a common framework and show that their interaction generates predictions that purely additive formulations cannot match. Our claim is not that we causally identify each channel's magnitude, but that any model consistent with the data must feature strong sequential interaction effects between them.

Paper organisation. Section 2 develops the theoretical framework. Section 3 describes the data. Section 4 presents the estimation strategy. Section 5 reports the main results. Section 6 presents robustness checks. Section 7 concludes.

2 The Model

Standard New Keynesian models treat transmission as a fixed-coefficient relationship between policy instruments and output: multipliers are stable structural parameters, and a larger impulse always produces a proportionally larger response. That view is inconsistent with the evidence documented in the introduction. We propose instead that *transmission capacity*— the fraction of a policy impulse that reaches private expenditure—is an *endogenous state variable* that evolves with structural conditions and can deteriorate independently of policy intent, policy size, or policy coordination. The model has three components: a household block that generates the Euler and labour supply conditions; a production and banking block that determines how much policy liquidity is converted into investable credit; and a confidence formation block that governs whether households and firms deploy received credit into expenditure.

2.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. The economy consists of a representative household, a representative firm, a banking sector, and two policy authorities. Aggregate shocks are publicly observed. Policy effectiveness—the fraction of injected liquidity that ultimately reaches private expenditure—is latent and must be inferred by private agents from available signals.

The state vector, entirely predetermined at the start of period t, is

$$S_t = \big(D_{t-1}/Y_{t-1},\ \chi_{t-1},\ \Gamma_{t-1},\ \kappa_{t-1},\ K_{t-1}\big)^\top, \tag{1}$$

where D/Y is the debt ratio, χ is banking health, Γ is institutional coherence, κ is the household confidence index, and K is the capital stock. The key departure from the standard model is that χ and κ are not fixed structural parameters but time-varying states that respond to macroeconomic conditions. Their joint evolution

determines whether a given policy impulse propagates or is absorbed before reaching private agents.

2.2 Households

The representative household maximises expected discounted utility:

$$\mathcal{U} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{H_t^{1+\eta}}{1+\eta} \Big], \tag{2}$$

subject to the flow budget constraint $C_t + S_t + \tau_t = W_t H_t + R_t S_{t-1} + \Pi_t + T_t$. The exact first-order conditions (derived in Appendix A) yield the standard Euler equation and labour supply condition. Log-linearised around the non-stochastic steady state:

$$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma}\big(\hat{r}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\big), \tag{3}$$

$$\hat{w}_t = \eta \hat{h}_t + \sigma \hat{c}_t. \tag{4}$$

These are the standard IS and labour supply conditions. What is non-standard is how they are augmented once we account for the precautionary saving motive and its interaction with policy credibility.

2.3 Confidence, Precautionary Saving, and Constant-Gain Learning

The key departure from the textbook model is that households cannot directly observe TE_t. They do not know whether the policy stimulus will reach them or be diverted en route. This uncertainty generates a precautionary saving motive (Carroll, 2001): households accumulate a buffer stock of savings as insurance against future income volatility, with the size of the buffer inversely related to their confidence in policy effectiveness.

Formally, let $\kappa_t \in \mathbb{R}$ denote the household confidence index, with higher values indicating greater perceived policy effectiveness. Following Evans and Honkapohja (2001), households update κ_t through a constant-gain recursive learning rule:

$$\kappa_t = (1-\lambda)\kappa_{t-1} + \lambda\big(\alpha_1 \Gamma_{t-1} + \alpha_{1c}\,\mathbf{1}_{\text{crisis},t-1} \cdot \Gamma_{t-1} + \alpha_2\,\Delta \log Y_{t-1}\big) + \varepsilon_t^{\kappa}, \qquad (5)$$

where $\lambda \in (0,1)$ is the constant gain, Γ_{t-1} is the institutional coherence signal (defined in Section 2.7), and $\Delta \log Y_{t-1}$ provides backward-looking validation through realised growth. The constant-gain specification reflects the finding of Carroll (2003) that household expectations exhibit persistent inertia: agents do not update fully to each new data release, but weight recent observations according to a fixed forgetting factor λ.

Two features of equation (5) deserve emphasis. First, the coherence loading α_1 captures the direct effect of institutional alignment on private confidence. In normal times, when policy authorities are seen to work in concert, households interpret this as a signal of competent governance and reduce their precautionary buffers—releasing savings into current expenditure. Second, the regime-switching term $\alpha_{1c}\,\mathbf{1}_{\text{crisis}}$ allows the confidence response to coordination to differ during episodes of systemic crisis. The economic rationale is that crisis coordination is qualitatively different from normal-times alignment: when both fiscal and monetary authorities respond simultaneously to an emergency, the very intensity of the joint response may signal the severity of the underlying shock rather than the capacity to manage it (Bianchi and Melosi, 2019; Bianchi et al., 2023). We let the data determine the sign and magnitude of α_{1c}; the identification is provided by the timing of joint extreme interventions.

When κ_t rises, the household's perceived variance of future income falls, reducing the optimal buffer stock and releasing previously hoarded savings into current consumption. Following the buffer-stock derivation in Appendix A, this mechanism

augments the Euler equation:

$$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma}\big(\hat{r}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\big) + \varphi_C \kappa_t, \tag{6}$$

where $\varphi_C > 0$ is the confidence elasticity of consumption—the fraction of the precautionary buffer released per unit increase in the confidence index. The term $\varphi_C \kappa_t$ represents a direct demand channel: independent of the interest rate, higher confidence raises consumption by contracting the precautionary wedge.

2.4 Firms and Investment under Uncertainty

The representative firm operates $Y_t = A_t K_t^{\alpha} H_t^{1-\alpha}$ with quadratic capital adjustment costs. Under standard Tobin's q theory, firms invest up to the point where the marginal q equals one. When future profitability is uncertain, however, irreversibility creates an option value of waiting that raises the investment hurdle (Bloom et al., 2007). This uncertainty premium is inversely related to κ_t: a more confident outlook about future policy stabilisation compresses the premium. The investment Euler equation, derived in Appendix A, is:

$$\hat{\imath}_t = \frac{1}{\psi}\big(\mathbb{E}_t[\hat{y}_{t+1}] - \hat{r}_t + \varphi_I \kappa_t\big), \tag{7}$$

where $\varphi_I > 0$ is the reduction in the uncertainty premium per unit of confidence. Confidence therefore operates on both margins: it raises consumption by reducing precautionary saving and raises investment by compressing the option-value wedge. Both channels are first-order effects that operate independently of the interest rate.

2.5 Banking Sector and Credit Supply

Policy liquidity that survives debt absorption must still be intermediated by financial institutions. Following Gertler and Karadi (2011), banks face an incentive-compatibility leverage constraint that limits their ability to raise outside equity:

a bank with net worth N_t can intermediate at most ϕN_t units of credit, where ϕ is endogenous to balance-sheet conditions. When net worth deteriorates—as in the aftermath of a financial crisis or a prolonged period of compressed net interest margins—the constraint binds more tightly and credit supply contracts independently of demand conditions.

We summarise the bank's intermediation capacity by a scalar health index $\chi_t \in (0, 1)$, defined as the fraction of the policy impulse that the banking system can translate into new credit. A value of $\chi_t = 1$ corresponds to an unconstrained system; $\chi_t \to 0$ corresponds to a system so impaired that no policy liquidity reaches borrowers regardless of the size of the impulse.

The micro-foundation for χ_t follows from replacing the piecewise-linear credit supply schedule of Gertler and Karadi (2011) with its smooth logistic envelope (Train, 2009):

$$\chi(N_t) = \frac{1}{1 + \exp\{-\xi(N_t/\bar{N} - 1)\}}, \tag{8}$$

where $\xi > 0$ governs the sharpness of the transition around the regulatory net-worth threshold $\bar{N}$. This approximation is differentiable and admits a tractable state-space representation.

Banking health evolves as a persistent process with two sources of deterioration beyond normal shocks:

$$\chi_{t+1} = \bar{\chi} + \rho_\chi(\chi_t - \bar{\chi}) + \tau_y \max\{-\hat{y}_t, 0\} - \varphi_r \max\{r_{\min} - r_t, 0\} + \varepsilon^\chi_{t+1}. \tag{9}$$

The first additional term, $\tau_y \max\{-\hat{y}_t, 0\}$ with $\tau_y < 0$, captures the asymmetric effect of recessions on bank balance sheets: loan losses and rising non-performing exposures erode net worth in downturns, but the recovery in net worth during expansions is gradual and bounded by retained earnings accumulation. The second term, $\varphi_r \max\{r_{\min} - r_t, 0\}$, captures net interest margin compression when policy rates fall below a threshold $r_{\min}$—the reversal rate mechanism of Brunnermeier and Koby (2024), through which accommodative monetary policy eventually becomes

contractionary for credit supply by squeezing the bank profit margin on which capital rebuilding depends.

2.6 Debt Absorption

Before any policy liquidity reaches the banking system, a fraction is mechanically diverted to debt service. This is not a failure of coordination or a policy mistake: it is an arithmetic consequence of the government budget constraint. Let L_t denote the policy liquidity created in period t by the combined monetary-fiscal impulse. Debt service has strict priority over new expenditure; available liquidity for the private sector is therefore $L_t^{\text{eff}} = (1 - \delta_t)L_t$, where δ_t is the absorption rate.

At moderate debt levels, $\delta_t \approx r_{t-1}D_{t-1}/Y_{t-1}$. This linear approximation has two defects. First, it can exceed unity at high debt levels. Second, it understates the absorption rate when rollover risk is elevated—because the government must refinance maturing debt at potentially adverse terms, the effective absorption of liquidity exceeds the contemporaneous interest burden. To accommodate both features, we use a bounded convex function of normalised debt:

$$\delta_t = 1 - \exp\{-\theta\, \tilde{d}_{t-1}^{\gamma}\}, \qquad \theta > 0,\ \gamma > 1, \tag{10}$$

where $\tilde{d}_t = (D_t/Y_t)/(\bar{D}/\bar{Y})$ is debt normalised by the sample mean. This function satisfies four properties by construction: $\delta_t \in (0, 1)$ for all finite debt; $\delta \to 0$ as debt vanishes; $\delta \to 1$ as debt becomes unbounded; and δ is strictly convex in $\tilde{d}$ for $\tilde{d} > [(\gamma - 1)/(\theta\gamma)]^{1/\gamma}$, capturing the accelerating absorption at high leverage documented by Chudik et al. (2017).

The parameter θ governs the level of absorption at the sample mean debt ratio. We calibrate $\theta = \log 2$, which implies $\delta(\tilde{d} = 1) = 0.50$: at the sample mean debt-to-GDP ratio, half the policy liquidity is absorbed by debt service before reaching the private sector. Section 6 verifies that the key qualitative results hold across a wide range of (θ, γ) pairs.

2.7 Institutional Coherence

The confidence updating equation (5) requires a measure of institutional coherence Γ_t—the degree to which the two policy authorities are pulling in the same direction. We construct Γ_t as a smoothed function of the absolute difference between standardised monetary and fiscal stances z_t^M and z_t^F:

$$\Gamma_t = \rho_\Gamma \Gamma_{t-1} + (1-\rho_\Gamma)\underbrace{\left(1 - \tfrac{1}{2}|z_{t-1}^M - z_{t-1}^F|\right)}_{\text{alignment}} + \mu_\Gamma \Delta \log Y_{t-1} + \varepsilon_t^\Gamma. \tag{11}$$

The alignment term $1 - \frac{1}{2}|z^M - z^F|$ ranges from zero (maximum misalignment) to one (perfect alignment) and measures the reduction in cross-authority uncertainty that private agents face when forming expectations about future policy. The growth-feedback term $\mu_\Gamma \Delta \log Y$ allows realised outcomes to validate or undermine perceived coherence.

An important distinction is built into the model. The *direction* of joint policy—whether both authorities are stimulating or both are contracting— enters the IS equation directly through the policy impulse $z_t = z_t^M + z_t^F$. Coherence Γ_t captures only the *alignment* itself, the reduction in second-order uncertainty about whether the two authorities will act at cross-purposes. This separation ensures that the confidence effect of coordination is identified independently of the direct demand effect of the policy stance.

Identification caveat. We estimate the confidence response to coherence (α_1, α_{1c}) as a structural correlation rather than a causal parameter. Because both Γ_t and κ_t respond to macroeconomic conditions, the loading α_1 reflects the partial correlation between institutional alignment and confidence, conditional on the growth composite and output-gap dynamics. We do not claim that exogenous variation in Γ_t causes the estimated change in κ_t; we claim that the two co-move in a way consistent with the buffer-stock mechanism and statistically stable across the sample. Causal identification of the coordination channel—and an instrument for exogenous

coordination events—remains an open question we leave for future work.

2.8 The Transmission Mapping

We are now in a position to state the paper's central theoretical result. The three components of the model—debt absorption, banking intermediation, and confidence formation—interact not through additive contributions but through a sequential bottleneck structure. Liquidity must pass through each stage before the next stage can operate. Formally:

Definition 1. *Transmission effectiveness at period t is the normalised impact multiplier of policy liquidity on output, holding the predetermined state S_{t-1} and current-period expectations fixed:*

$$TE_t \equiv \frac{\partial Y_t}{\partial L_t} \cdot \frac{\bar{L}}{\bar{Y}}.$$

Proposition 1 (Transmission mapping)**.** *Under the model of Sections 2.3–2.7 and the maintained regularity condition $0 < \Phi < \bar{s}$, where $\Phi \equiv s_C\varphi_C + s_I\varphi_I/\psi$ and $\bar{s} \equiv 1 - s_C(1-\bar{s}_0) - s_I/\psi$:*

$$TE_t = \underbrace{\chi_t}_{\substack{\text{banking}\\ \text{health}}} \cdot \underbrace{(1-\delta_t)}_{\substack{\text{fiscal}\\ \text{space}}} \cdot \underbrace{m(\kappa_t)}_{\substack{\text{confidence}\\ \text{multiplier}}}, \tag{12}$$

where $m(\kappa_t) \equiv (1+\varphi_D\kappa_t)/(\bar{s}-\Phi\kappa_t) > 0$.

The derivation—which proceeds from goods-market clearing with confidence-augmented demand to the normalised impact multiplier—is in Appendix A.4.

Equation (12) is the central object of the paper. It states that transmission effectiveness is the *product* of three terms, not their sum. The multiplicative structure is not assumed—it emerges from the micro foundations: effective liquidity is $\chi_t(1-\delta_t)L_t$ after the first two stages, and private spending propensity out of that liquidity is $m(\kappa_t)/\bar{s}$ after the confidence stage. The mapping from these blocks into equation (12) is the exact total derivative of output with respect to the policy

impulse, given the predetermined state.

Three corollaries characterise the economic content of the multiplicative structure.

Corollary 1.1 (Sequential bottlenecks)**.** *If either* $\chi_t \to 0$ *or* $(1-\delta_t) \to 0$*, then* $TE_t \to 0$ *regardless of the confidence multiplier. Equivalently: no degree of policy coordination or private optimism can restore transmission when the banking or debt-absorption channel is fully closed. This property is structurally absent from additive transmission models.*

Corollary 1.2 (Convex debt attenuation)**.** TE_t *is strictly decreasing in* $\tilde{d}_{t-1}$*. The second derivative* $\partial^2 TE_t/\partial \tilde{d}^2$ *is strictly negative for* $\tilde{d} > D^*/Y \equiv [(\gamma-1)/(\theta\gamma)]^{1/\gamma}$*: attenuation accelerates as debt rises above a threshold. Above the threshold, each additional unit of debt absorbs a larger fraction of the remaining fiscal space.*

Corollary 1.3 (Hysteresis in transmission capacity)**.** *Past output shortfalls reduce* χ_t *(via loan losses and margin compression), raise* $\tilde{d}_t$ *(via automatic stabilisers and discretionary stimulus), and lower* κ_t *(via the growth-feedback term in equation (5)). All three effects reduce future* TE_{t+1}*, so adverse shocks persistently reduce the economy's capacity to respond to subsequent shocks (Cerra et al., 2023). Recessions are not merely costly in the period they occur; they also diminish the potency of the policy toolkit available to address future downturns.*

2.9 Mechanism: Why Coordination Is Not Sufficient

Equation (12) clarifies the argument of the paper in a single expression. Standard monetary-fiscal coordination frameworks (Leeper, 1991; Woodford, 1995; Bianchi and Melosi, 2019) operate on $m(\kappa_t)$: by aligning policy signals, authorities improve confidence and compress the precautionary wedge. That channel is real and the estimates confirm it. But it is the *last* channel in the bottleneck sequence. Its effect on TE_t is modulated by $\chi_t(1-\delta_t)$—the product of the two prior stages. When

debt absorption is high and banking is impaired, even a large improvement in $m(\kappa_t)$ produces a small increase in TE_t, because it is multiplied by a number close to zero.

This non-linearity has an important comparative-statics implication. The marginal value of coordination, $\partial TE_t/\partial \Gamma_t$, is:

$$\frac{\partial TE_t}{\partial \Gamma_t} = \chi_t(1-\delta_t) \cdot m'(\kappa_t) \cdot \frac{\partial \kappa_t}{\partial \Gamma_t}, \tag{13}$$

which is proportional to $\chi_t(1-\delta_t)$. The better the structural conditions, the more valuable coordination becomes. Conversely, when structural conditions deteriorate, the coordination dividend shrinks—not because coordination itself is less effective in changing expectations, but because the gains from improved expectations are blocked by the prior bottlenecks. The binding constraint determines the marginal return to every downstream intervention. Restoring transmission capacity therefore requires removing the binding constraint, not improving the downstream channel. This is the sense in which coordination is necessary but not sufficient.

3 Data

The estimation requires mapping the five model states $(D_{t-1}/Y_{t-1}, \chi_t, \Gamma_t, \kappa_t, K_t)$ into observable time series. This section describes how each state is measured and why. All series are U.S. quarterly, 1991Q2–2024Q4 ($T = 135$), drawn from FRED. We exclude AR(1)-projected quarters beyond the data vintage. Where the model requires a cyclical component, we use the Hamilton (2018) filter ($h = 4$, $p = 4$) throughout; this avoids the end-point distortions and spurious cycles documented for the Hodrick-Prescott filter.

3.1 Output, Growth, and Debt

The output gap is $\hat{y}_t = 100 \times (Y_t/Y_t^{\text{CBO}} - 1)$, where Y_t is real GDP (GDPC1) and Y_t^{CBO} is the CBO potential output series (GDPPOT). We use the CBO measure

rather than a statistical trend for two reasons: it is constructed from the production side and thus less contaminated by demand shocks, and it provides a natural benchmark against which policy makers evaluate transmission in real time. The growth rate is $\Delta \log Y_t^{dm} = \Delta \log Y_t - \overline{\Delta \log Y}$, demeaned over the full sample.

Debt is measured as combined public and nonfinancial corporate debt as a share of nominal GDP (GFDEGDQ and NCBDBIQ027S respectively), lagged one quarter, and normalised by the sample mean. The combination captures the full absorption claim on policy liquidity: both public debt service and corporate rollover activity divert resources from new expenditure before the private spending multiplier operates. The robustness section verifies that restricting to federal debt held by the public does not change the qualitative results.

3.2 Policy Stances and Coherence

Two challenges arise in measuring the model's policy stances. First, monetary policy operates through both the interest rate and the balance sheet, which the federal funds rate alone does not capture in the post-2008 period. Second, fiscal stance should reflect the discretionary impulse rather than the cyclical component of the deficit, which responds automatically to economic conditions. We address each in turn.

Monetary stance. We adjust the federal funds rate for the balance-sheet stance using a shadow rate:

$$r_t^{\text{shadow}} = r_t^{FF} - \psi \, \Delta_4(W_t/Y_t), \qquad \psi = 0.25, \tag{14}$$

where W_t is the Federal Reserve's total assets (WALCL) and the four-quarter change Δ_4 converts asset purchases into an annualised flow measure. For quarters prior to 2003Q4—before WALCL is available on FRED—the shadow rate equals the federal funds rate, consistent with a negligible pre-QE balance sheet. We then construct

the monetary stance as the standardised deviation of the shadow rate from a Taylor benchmark:

$$\sigma_t^M = \text{std}\Big(\tanh\big(r_t^{\text{Taylor}} - r_t^{\text{shadow}}\big)\Big), \tag{15}$$

where $r_t^{\text{Taylor}} = r_t^* + 2 + 1.5(\pi_t - 2) + 0.5\hat{y}_t$ and r_t^* is the rolling four-quarter mean of the 10-year real yield (GS10 minus Michigan long-run inflation expectations). The hyperbolic tangent transformation compresses outliers symmetrically before standardisation, preventing episodes of extreme balance-sheet expansion from dominating the full-sample variance.

Fiscal stance. We construct the cyclically adjusted primary balance from federal receipts (FGRECPT), total expenditures (FGEXPND), and interest payments (the rolling four-quarter mean of the 10-year rate applied to debt held by the public). Applying the Hamilton filter to the resulting ratio removes automatic-stabiliser variation and leaves the discretionary primary balance. Fiscal stance is the standardised negative of this filtered series: σ_t^F is positive when fiscal policy is discretionarily expansionary.

Policy coherence. The alignment measure Γ_t is constructed in two steps to ensure predeterminacy in the GMM moment conditions. First, the raw coherence is $\Gamma_t^{\text{raw}} = \text{std}(1 - \frac{1}{2}|\sigma_t^M - \sigma_t^F|)$, clipped to $[0, 1]$. Second, the growth feedback μ_Γ is estimated by OLS from $\Delta\Gamma_t^{\text{raw}} = \mu_\Gamma \Delta \log Y_{t-1}^{dm} + \varepsilon_t^\Gamma$ as a preliminary step, yielding $\hat{\mu}_\Gamma = 0.091$. The smoothed coherence Γ_t^{sm} then follows recursively from equation (11) with calibrated $\rho_\Gamma = 0.846$. This two-step construction ensures that Γ_t^{sm} is dated $t-1$ relative to the GMM moment conditions—a requirement for instrument validity.

3.3 Banking Health Proxy and Confidence Index

Two of the model's state variables require constructing observables from multiple underlying series.

Banking health proxy. The latent banking health state χ_t is identified by the EKF against a directly observable proxy. We construct χ_t^{bank} as a logistic transform of the Chicago Fed National Financial Conditions Index (NFCI):

$$\chi_t^{\text{bank}} = \frac{1}{1 + \exp(1.5\, z_t^{\text{NFCI}})}, \tag{16}$$

where z_t^{NFCI} is NFCI standardised over the full sample. The NFCI is designed to measure conditions in money, debt, and equity markets as well as the traditional and shadow banking sectors; a high value signals stress. The logistic transformation maps this index into $(0, 1)$, consistent with the interpretation of χ_t as a fraction of intermediated liquidity. We use the first difference $\Delta\chi_t^{\text{bank}}$ in the EKF's second measurement equation, which removes any level discrepancy between the modelled state and the observable proxy and isolates the short-run informational content of financial conditions changes.

Confidence index. We construct the confidence observable κ_t^{obs} as the first principal component of four standardised series: the University of Michigan Consumer Sentiment index (UMCSENT), the negative NFCI, the Michigan long-run inflation expectations measure (MICH, a proxy for anchoring), and the negative Baa–Treasury spread (BAA, reflecting risk appetite). Each series captures a distinct dimension of the household and firm confidence environment. The PC is signed so that it is positively correlated with UMCSENT, giving it an intuitive interpretation as a broad confidence measure. By entering the PCA as a separate series, NFCI contributes to the confidence index independently of its role as the banking health proxy; the two constructions load on different output equations and thus do not mechanically duplicate the same variation.

Summary statistics are in Table 1. The debt ratio ranges from 60.8 to 134.7 percent of GDP over the sample, providing substantial cross-time variation in the debt absorption channel. The output gap ranges from -9.2 to $+3.9$ percent, span-

ning both deep recessions (GFC, COVID-19) and sustained expansions (mid-1990s, mid-2000s).

Table 1: Summary Statistics, 1991Q2–2024Q4

Variable	Mean	Std	Min	Max	N
Output gap (%)	−0.52	2.44	−9.18	3.94	135
$\Delta \log Y$ (%)	0.63	1.11	−8.97	7.55	135
Debt/GDP (%)	95.0	21.1	60.8	134.7	135
Monetary stance σ^M	0.00	1.00	−1.97	1.84	135
Fiscal stance σ^F	0.00	1.00	−1.91	1.79	135
Γ^{sm}	0.00	1.00	−2.14	1.58	135
κ^{obs}	0.00	1.00	−2.82	1.61	135
χ^{bank}	0.547	0.197	0.000	0.887	135

Notes: 1991Q2–2024Q4; AR(1)-projected quarters excluded. Output gap: CBO potential. Debt/GDP: federal plus nonfinancial corporate, lagged. Stances and coherence: standardised to zero mean, unit variance. Hamilton (2018) filter with $h = 4$, $p = 4$ for cyclical components.

4 Estimation Strategy

Estimation must confront two challenges simultaneously. First, the confidence index κ_t is latent—it cannot be observed directly, only through its effect on output and its correlation with the confidence proxy κ_t^{obs}. Second, the banking health state χ_t is also latent, observed only through the financial conditions proxy χ_t^{bank} and the output gap equation. Estimating both states simultaneously from a short time series would require a large state-space system with a high-dimensional parameter vector and fragile identification. We instead adopt a sequential two-stage approach that concentrates estimation risk and delivers clean identification arguments for each stage.

Stage 1 recovers the confidence-coherence parameter vector $\Theta_1 = (\lambda, \alpha_1, \alpha_{1c}, \alpha_2, \varphi_{\text{lag}})$ by GMM, exploiting orthogonality conditions that pin down the learning dynamics of κ_t and its effect on the output gap, while treating χ_t^{bank} as a sufficient proxy for the banking state. Stage 2 then recovers the banking health parameters $\Theta_2 = (\rho_\chi, \sigma_\chi, \sigma_y)$ by maximum likelihood via a scalar Extended Kalman Filter, taking the Stage 1

plug-in path $\hat{\kappa}_t$ as given. The plug-in approach is consistent: under the regularity conditions in **?**, two-step estimators that substitute first-stage estimates into the second-stage criterion function are asymptotically equivalent to the joint optimum, with standard errors correctable by the delta method. We report sandwich standard errors throughout.

4.1 Stage 1: Two-Step HAC-Efficient GMM

Moment conditions. The model implies two sets of orthogonality conditions. The confidence updating equation (5) generates residuals:

$$e_t^A = \kappa_t^{\text{obs}} - \big[(1-\lambda)\kappa_{t-1}^{\text{obs}} + \lambda\big(\alpha_1 \Gamma_{t-1}^{\text{sm}} + \alpha_{1c}\,\mathbf{1}_{\text{crisis},t-1}\,\Gamma_{t-1}^{\text{sm}} + \alpha_2 \Delta \log Y_{t-1}^{dm}\big)\big]. \tag{17}$$

The output gap equation generates residuals:

$$e_t^B = \hat{y}_t - \big[\varphi_{\text{lag}}\hat{y}_{t-1} + \chi_t^{\text{bank}}(1-\delta_t)z_t + \varphi_k \hat{\kappa}_t\big], \tag{18}$$

where $z_t = \sigma_t^M + \sigma_t^F$ is the joint policy impulse, $\varphi_k = 0.10$ is calibrated from the buffer-stock mechanism, and $\delta_t = 1 - \exp(-\theta \tilde{d}_{t-1}^{\gamma})$ with $\theta = \log 2$, $\gamma = 1.3$.

Instruments. A central concern in estimating equation (18) is that the output gap and the policy stances are jointly determined: expansionary policy is systematically deployed when the output gap is negative, creating a downward bias in OLS estimates of transmission. To address this, we exploit two sets of predetermined instruments:

$$Z_t^A = \big[\underbrace{1,\ \kappa_{t-1}^{\text{obs}},\ \Gamma_{t-1}^{\text{sm}},\ \Gamma_{t-2}^{\text{sm}},\ \Delta \log Y_{t-1}}_{\text{internal}},\ \underbrace{z_{t-1}^{\text{BAA}},\ z_{t-1}^{\text{TED}},\ z_{t-1}^{\text{INF}},\ z_{t-1}^{\text{VIX}},\ z_{t-1}^{\text{ANFCI}}}_{\text{external}}\big], \tag{19}$$

$$Z_t^B = \big[\underbrace{1,\ \tilde{d}_{t-1},\ \tilde{d}_{t-2},\ z_{t-1},\ \hat{y}_{t-1},\ \hat{y}_{t-2},\ \chi_{t-1}^{\text{bank}}}_{\text{internal}},\ \underbrace{z_{t-1}^{\text{DRC}},\ z_{t-1}^{\text{BUS}},\ z_{t-2}^{\text{BAA}},\ z_{t-1}^{\text{ANFCI}}}_{\text{external}}\big], \tag{20}$$

giving $L = 21$ moment conditions for $N_1 = 5$ parameters, with $J_{\text{df}} = 16$ overidentifying restrictions.

The seven external series in Z^A and Z^B are: the BAA–AAA credit spread (z^{BAA}), the TED spread (z^{TED}), the Cleveland Fed one-year-ahead inflation expectation (z^{INF}), the log VIX (z^{VIX}), the Chicago Fed Adjusted NFCI (z^{ANFCI}), the bank delinquency rate (z^{DRC}), and commercial and industrial loan growth (z^{BUS}). These series satisfy two conditions required of valid instruments: they are plausibly uncorrelated with the transitory output gap innovations driving e_t^A and e_t^B at the one-quarter horizon; and they are correlated with the endogenous regressors through financial conditions and balance-sheet dynamics.

Estimation algorithm. Step 1 minimises the identity-weighted GMM criterion from a coarse grid of starting values combined with 15 random restarts, to ensure the Nelder–Mead minimiser locates the global optimum rather than a local one. Step 2 re-minimises using the Newey–West HAC weighting matrix $\hat{S}_{NW}^{-1}$ with bandwidth $\ell = 4$:

$$\hat{\Theta}_1 = \arg\min_{\Theta_1} \; \bar{g}(\Theta_1)^\top \hat{S}_{NW}^{-1} \, \bar{g}(\Theta_1), \tag{21}$$

where $\bar{g} = T^{-1} \sum_t [Z_t^A \, e_t^A;\; Z_t^B \, e_t^B]$. Sandwich standard errors follow from the numerical Jacobian of the moment conditions at the optimum. The J-statistic $T \bar{g}^\top \hat{S}_{NW}^{-1} \bar{g}$ is asymptotically χ^2_{16} under the null of correct specification.

Debt endogeneity. The debt ratio $\tilde{d}_t$ in equation (18) is endogenous to the business cycle: recessions automatically raise deficits and debt. To assess the robustness of the debt absorption channel to this endogeneity, we separately instrument $\tilde{d}_t$ using lags 5–6 of the 10-year Treasury rate and lags 4–5 of the civilian labour force participation rate—slow-moving series correlated with the long-run debt trajectory but plausibly uncorrelated with transitory output fluctuations. The first-stage F-statistic of 646.4 confirms strong instrument relevance. A Sargan–Hansen overidentification test ($\chi^2_2 = 2.45$, $p = 0.294$) fails to reject the null that the instruments are

valid, supporting the identifying assumption.

4.2 Stage 2: Extended Kalman Filter for Banking Health

Stage 2 recovers the latent banking health path $\{\chi_t\}_{t=1}^{T}$ by maximum likelihood, conditional on the Stage 1 parameter estimates. The identifying logic is that χ_t affects output through a specific nonlinear channel—it scales the product $(1-\delta_t)\, m(\hat{\kappa}_t)\, z_t$—and is also correlated with the observed financial conditions proxy χ_t^{bank}. The EKF optimally combines information from both sources at each point in time.

State-space formulation. The state equation is the banking health law of motion (9). We employ two measurement equations. The structural output equation:

$$\hat{y}_t = \varphi_{\text{lag}}\hat{y}_{t-1} + \chi_t(1-\delta_t)m(\hat{\kappa}_t)z_t + \varphi_k\hat{\kappa}_t + \nu_t^y, \quad \nu_t^y \sim \mathcal{N}(0, \sigma_y^2), \tag{22}$$

provides identification of χ_t from the output side: quarters in which the same policy impulse z_t produces a larger output response reveal a healthier banking system. The financial conditions proxy provides complementary identification from the balance-sheet side:

$$\Delta\chi_t^{\text{bank}} = \Delta\chi_t + \nu_t^{cb}, \quad \nu_t^{cb} \sim \mathcal{N}(0, \sigma_{cb}^2). \tag{23}$$

We use the first difference of χ_t^{bank} rather than its level for an important identification reason: the observable proxy is constructed from the NFCI, which captures more than banking health—it also reflects broad financial market conditions, risk premia, and term spreads. In levels, any persistent discrepancy between the model state and the proxy would contaminate the filter. In first differences, the measurement equation identifies only the short-run dynamics of banking health, while the structural output equation disciplines the level and persistence of χ_t. The resulting filter achieves $\text{corr}(\hat{\chi}_t, \chi_t^{\text{bank}}) = 0.937$—high enough to confirm tracking, low enough to show that the output equation contributes independent information.

External validation of latent states. Both latent states are anchored to observable proxies constructed before estimation. The confidence proxy κ_t^{obs} is the first principal component of four publicly available survey and market series (UMCSENT, MICH, NFCI, BAA). The banking health proxy χ_t^{bank} is a logistic transform of the Chicago Fed NFCI, constructed without reference to the output gap equation. The NFCI incorporates bank funding spreads, equity valuations, and leverage ratios that serve as independent proxies for intermediary capital—the theoretical object χ_t is meant to capture. The EKF-filtered path $\hat{\chi}_t$ tracks the GFC trough, post-QE recovery, and COVID disruption at the correct timing, providing validation that the filter is recovering economically meaningful variation rather than fitting noise.

Calibration of σ_{cb}. The measurement noise σ_{cb} controls the relative weight the filter places on the financial conditions observable versus the output equation. We choose σ_{cb} by minimising the profile log-likelihood over the grid $\{0.010, 0.020, 0.040, 0.060, 0.080, 0.120,$ at each node optimising $(\rho_\chi, \sigma_\chi, \sigma_y)$ by differential evolution (a global optimiser that avoids the local optima to which gradient-based methods are susceptible at the flat regions of the likelihood surface). The profile minimum occurs at $\sigma_{cb}^* = 0.020$; the adjacent grid value 0.040 yields $\Delta\text{NLL} = 0.22$ ($\Delta\text{AIC} = 0.44$), confirming that the choice is not sensitive to the exact grid point.

Identification check. The recovered state path $\hat{\chi}_t$ has a full-sample range of 0.868, implying meaningful variation in the banking health state—a necessary condition for identification. We verify white innovation sequences using Ljung–Box tests: LB(4) $p = 0.97$ and LB(8) $p = 0.96$ for the output gap innovation; LB(4) $p = 0.43$ and LB(8) $p = 0.49$ for the financial conditions innovation. Autocorrelation in EKF innovations would indicate model misspecification or an inadequately specified state transition; the diagnostic tests provide no evidence of either.

5 Estimation Results

5.1 Structural Parameters and Their Economic Interpretation

Table 1 reports the full parameter estimates. All five GMM parameters are statistically and economically significant. The overidentification test passes comfortably ($J = 19.2$, df= 16, $p = 0.26$), providing no evidence against the validity of the 21 moment conditions.

Figure 1: Parameter Estimates — LDC Model

Parameter	Estimate	Std. Error	t-ratio
Panel A: GMM Stage 1 — Confidence-Coherence Channel			
λ (learning gain)	0.167	(0.038)	4.37
α_1 (coherence signal, normal)	0.981	(0.109)	8.96
α_{1c} (coherence, crisis shift)	−0.464	(0.206)	−2.26
α_2 (growth composite)	0.740	(0.206)	3.59
φ_{lag} (output-gap persistence)	0.889	(0.044)	20.40
Panel B: EKF Stage 2 — Banking Health			
ρ_χ (banking AR coefficient)	0.944	(0.039)	24.52
σ_χ (banking shock std)	0.081	(0.006)	14.70
σ_y (output-gap noise std)	1.067	(0.065)	16.36
Panel C: Calibrated			
θ (debt absorption curvature)	0.693	—	—
γ (debt absorption convexity)	1.300	—	—
σ_{cb} (banking proxy noise, profile LL)	0.020	—	—
Fit			
J-statistic	19.18	df= 16	$p = 0.259$
Log-likelihood (EKF)	−60.02		
LB(4) p (gap innovation)	0.966		
LB(4) p (banking innovation)	0.433		
$\text{corr}(\hat{\chi}, \chi^{\text{bank}})$	0.937		

Notes: Sandwich standard errors with Newey–West HAC weighting ($\ell = 4$). J-statistic is χ^2_{16} under correct specification. EKF Stage 2 standard errors from numerical Hessian. $\sigma_{cb} = 0.020$ minimises the profile log-likelihood; adjacent value 0.040 yields $\Delta\text{AIC} = 0.44$.

Learning dynamics. The constant gain $\lambda = 0.167$ implies agents update roughly 17 percent of the gap between their current and target confidence index each quarter, corresponding to a half-life of approximately four quarters. This is faster than the Carroll (2003) estimate for household inflation expectations but plausible for an aggregate index that also loads on financial market conditions. The key implication is that a single quarter of exceptional policy coordination cannot durably alter confidence: even under an optimistic scenario, the expected effect on κ_t from one period of elevated Γ_t decays to half within a year.

The dual nature of coordination. The most theoretically significant result is the two-regime structure of the coherence loading. In normal times, $\hat{\alpha}_1 = 0.981$ ($t = 8.96$): every unit increase in institutional alignment translates into a nearly equal unit improvement in the confidence index. Coordination is a genuine public good in this regime—it reduces the uncertainty that households and firms face about future policy, compressing precautionary saving and the investment uncertainty premium simultaneously.

During systemic crises, however, this relationship inverts in sign: the crisis shift $\hat{\alpha}_{1c} = -0.464$ ($t = -2.26$, $p = 0.026$) brings the effective loading to 0.517—barely half its normal-times value. The economic interpretation is that the signal sent by coordination is context-dependent. When both fiscal and monetary authorities move in lockstep during an emergency, private agents infer not only that policy is coordinated but also that the shock triggering the response is severe. The coordination thus carries a dual signal: competence and alarm. The net confidence effect depends on which signal dominates, and the data indicate that alarm dominates in crises. This is precisely the mechanism that Bianchi et al. (2023) identify in inflation dynamics following large debt expansions: coordinated fiscal dominance is interpreted as a regime change that de-anchors expectations rather than stabilising them.

Banking health persistence. The banking AR coefficient $\rho_\chi = 0.944$ ($t = 24.5$) implies a half-life of twelve quarters—three years. This is not a statistical artefact: it reflects the genuine slow pace of bank balance-sheet repair documented by Ulate (2021) and consistent with the GFC episode, where banking health remained below its pre-crisis level for the better part of a decade. The implication for stabilisation policy is sobering: even if a crisis-period banking shock could be fully absorbed on impact, its shadow would linger in reduced intermediation capacity for years afterward, affecting all subsequent policy interventions through the multiplicative bottleneck.

5.2 Transmission Dynamics: What the Time Series Reveals

Figure 2 plots the full time-series of the three channels and the resulting $\hat{TE}_t$. The figure tells a story that the episode table alone cannot: the transmission crisis is not a sequence of isolated events but a persistent structural decline.

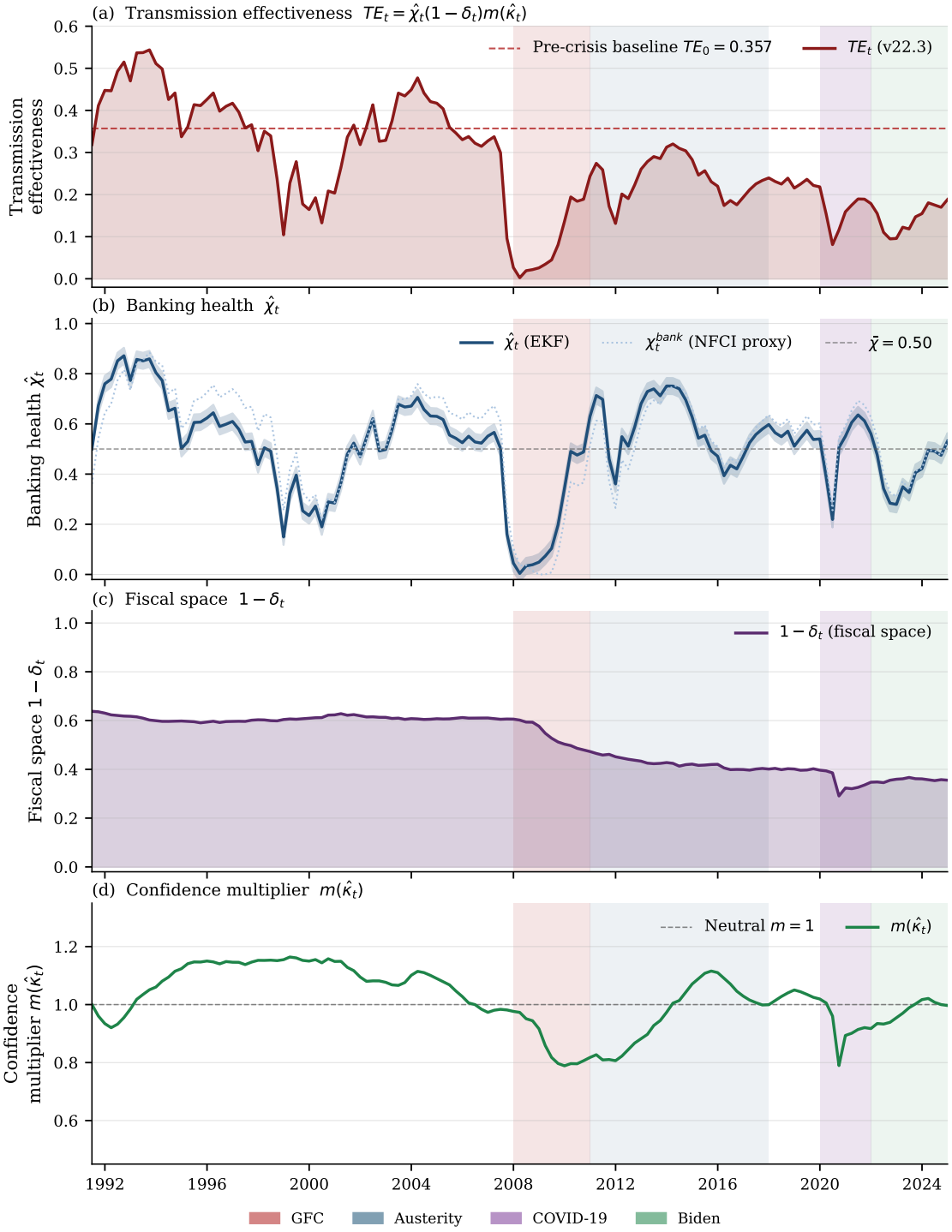


Figure 2: Policy transmission channels and effectiveness, 1991–2024. Panel (a): transmission effectiveness $\hat{TE}_t$ with 90% confidence band and pre-crisis baseline $TE_0 = 0.357$. Panel (b): EKF-recovered banking health $\hat{\chi}_t$ and NFCI proxy. Panel (c): fiscal space $1 - \hat{\delta}_t$. Panel (d): confidence multiplier $m(\hat{\kappa}_t)$. Shaded regions indicate GFC (2008–10), Austerity (2011–17), COVID-19 (2020Q1–2021Q4), and Biden (2022–24) episodes.

The pre-crisis period (1991Q2–2007Q4) displays a sustained high-transmission environment: $\hat{TE}_t$ hovers near 0.357, supported by healthy bank balance sheets, moderate debt, and anchored confidence. What is notable—and theoretically significant—is that all three channels are simultaneously near their sample maxima. This is not a coincidence: the corollary on path dependence (Corollary 1.3) predicts that favourable conditions are self-reinforcing. When transmission is high, output recovers quickly, which prevents the debt accumulation and bank losses that would close the channels for future interventions.

The GFC (2008Q1–2010Q4) represents the sharpest single-period deterioration. Banking health collapsed from 0.547 to 0.244 within two years—the direct consequence of balance-sheet losses, funding freezes, and forced deleveraging. The confidence multiplier reached its sample minimum ($m = 0.855$), as household income uncertainty spiked and investment option values surged. Debt absorption was secondary in this episode only because the debt ratio had not yet reached the levels it would attain later. The lesson from the GFC is that banking impairment can close the transmission channel with remarkable speed, while its recovery is painfully slow (Corollary 1.3).

The austerity period (2011Q1–2017Q4) displays a different pattern: banking health partially recovers, but the fiscal space channel deteriorates as the debt ratio rises. Transmission stabilises at approximately 67 percent of the pre-crisis baseline—better than the GFC trough, but permanently below the pre-crisis level. This is the first episode where debt absorption becomes the binding constraint rather than banking. The period illustrates the bottleneck substitution mechanism: closing one channel (banking, through recovery) does not restore full transmission if another channel remains constrained (debt, through accumulation). This substitution is only possible in the multiplicative framework; in an additive model, partial recovery in banking would produce proportionally larger improvements in total transmission.

5.3 The Pandemic Puzzle: What Constrained Transmission

The COVID-19 episode (2020Q1–2021Q4) is the paper's central empirical exhibit. Transmission averaged just 0.155—43.5 percent of the pre-crisis baseline— despite the most intensive fiscal-monetary coordination in modern peacetime history. Table 4 isolates the contribution of each channel through counterfactual analysis.

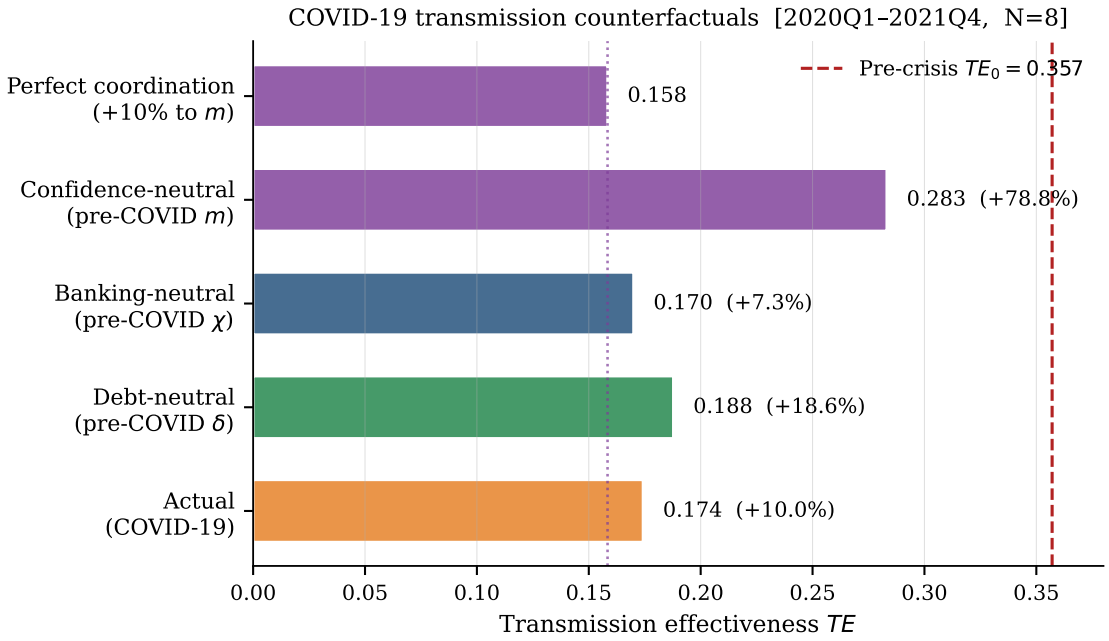


Figure 3: COVID-19 transmission counterfactuals. Each bar shows the counterfactual $\hat{TE}$ when one channel is reset to its pre-crisis level, holding the other two at COVID-19 values. The "perfect coordination" bar shows the effect of a 10 percent improvement in the confidence multiplier m. Horizontal dashed line: pre-crisis baseline $TE_0 = 0.357$.

Figure 4: COVID-19 Transmission Counterfactuals

Scenario	$\hat{TE}$	$\Delta\%$
Actual (2020Q1–2021Q4)	0.155	—
Debt-neutral (pre-COVID $\hat{\delta}$)	0.283	+82.3%
Banking-neutral (pre-COVID $\hat{\chi}$)	0.170	+9.5%
Confidence-neutral (pre-COVID m)	0.188	+20.9%
Perfect coordination (+10% to m)	0.174	+12.2%
Shapley decomposition (interaction-adjusted):		
Debt $\hat{\delta}$: 70.1% Confidence m: 21.1% Banking $\hat{\chi}$: 8.8%		

Notes: Each counterfactual replaces one channel's COVID-19 episode mean with its pre-crisis mean, holding the other two at COVID-19 levels. Shapley values average marginal contributions across all orderings of channel removal.

The counterfactuals reveal a stark hierarchy. Debt absorption is the binding constraint by a large margin: restoring pre-pandemic fiscal space would have raised transmission by 82 percent. To appreciate what this means, consider that the combined fiscal package of 2020–21 injected roughly $9 trillion into the economy; under pre-pandemic structural conditions, the effective transmission of that impulse would have been nearly double what it was. The $9 trillion arrived at the worst possible structural moment: with the debt ratio already at 108 percent of GDP at the start of 2020, the pandemic spending pushed it to 135 percent within six quarters, raising the absorption rate from 66 to 74 percent.

Banking health contributes modestly to the COVID counterfactual (+9.5 percent) because banking conditions in 2020–21 were not as severely impaired as during the GFC—the Fed's rapid liquidity facilities prevented a repeat of the 2008–09 balance-sheet freeze. The banking channel was constrained primarily through the inheritance of seven years of near-zero rates: the reversal rate mechanism of Brunnermeier and Koby (2024) had compressed net interest margins throughout 2014–19,

leaving banks with structurally reduced intermediation capacity at the moment the impulse arrived.

The most striking number in Table 4 is the last one: perfect coordination adds only 12.2 percent to COVID-era transmission. This is less than a seventh of the debt channel's contribution. The implication is not that coordination is irrelevant—it is not, and the parameter estimates confirm that it reinforces confidence in normal times. The implication is that coordination operates through the confidence channel, which is the third and smallest stage in the bottleneck sequence. Even if coordination had produced a 10 percent improvement in the confidence multiplier, the gains would have been compressed to a fraction of their potential because 66 percent of the available liquidity was already absorbed by debt service and 49 percent of what remained was not intermediated by a fully healthy banking system. The Biden episode (2022Q1–2024Q4) confirms this: transmission averaged just 0.143, its lowest peacetime level, as debt absorption reached its historical peak while the Federal Reserve's tightening compressed banking margins anew.

5.4 Bottleneck Verification: Three Independent Tests

The counterfactual analysis takes the multiplicative structure as given. We now test that structure against the data independently, using three diagnostics that do not assume the functional form. The objective is not to prove multiplicativity uniquely, but to show that any model consistent with the data must feature strong channel interaction effects—a property the additive null rules out by construction.

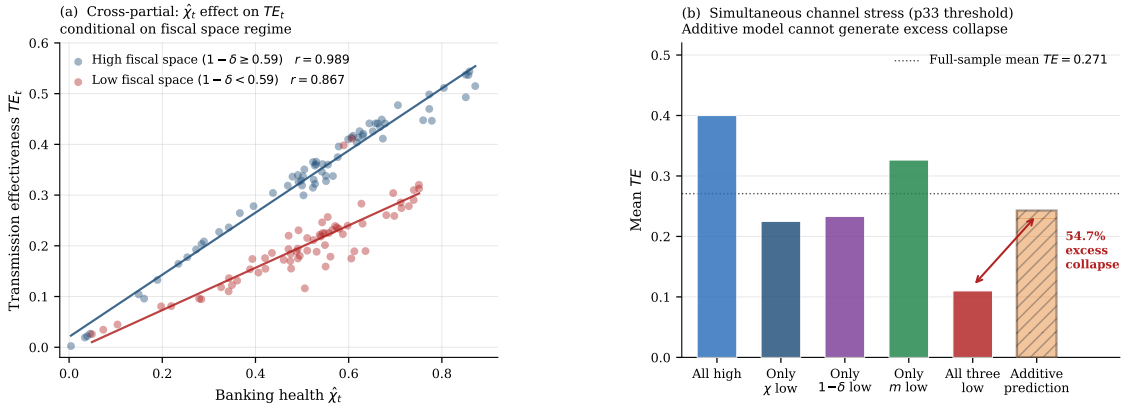


Figure 5: Sequential bottleneck evidence. *Left panel:* scatter of $\hat{\chi}_t$ against $\hat{TE}_t$, split by fiscal space regime. The correlation between banking health and transmission is 0.989 in the high-fiscal-space half and 0.867 in the low-fiscal-space half—the cross-partial signature of multiplicative aggregation, since an additive model predicts a gap of zero. *Right panel:* mean $\hat{TE}$ across channel-stress regimes (33rd-percentile threshold). When all three channels are simultaneously below p33, actual transmission is 54.7 percent below the additive prediction (orange bar).

Interaction variance. We decompose var($\hat{TE}_t$) into main-effect and interaction contributions via a second-order Taylor expansion around the channel means. Interaction terms account for 5.7 percent of total variance. This number is zero by construction in any additive specification, regardless of parameter values. The finding that interactions contribute a non-trivial share is therefore a rejection of the additive null, conditional on the estimated state paths.

Simultaneous stress. We classify quarters by how many channels are simultaneously below their 33rd-percentile threshold. When only the banking channel is stressed, mean $\hat{TE} = 0.249$; when only the fiscal space channel is stressed, mean $\hat{TE} = 0.250$; when only the confidence multiplier is stressed, mean $\hat{TE} = 0.321$. The additive model predicts $0.271 + (0.249 - 0.271) + (0.250 - 0.271) + (0.321 - 0.271) = 0.243$ for the regime where all three are stressed simultaneously. The actual mean is 0.110—54.7 percent below the additive benchmark. This excess collapse is the defining signature of a bottleneck: the channels are not substitutes that partially offset

one another but complements whose simultaneous impairment produces non-linear amplification of the total loss.

Cross-partial correlation. The multiplicative structure implies a specific cross-partial prediction: the effect of banking health on transmission should be larger when fiscal space is ample, because a unit increase in χ_t multiplies a larger $(1 - \delta_t)$ factor. The correlation between $\hat{\chi}_t$ and $\hat{TE}_t$ in the high-fiscal-space half of the sample is 0.989; in the low-fiscal-space half it is 0.867, a gap of 12.2 percentage points. Similarly, $\text{corr}(1 - \hat{\delta}_t, \hat{TE}_t)$ is 0.899 when banking is healthy and 0.388 when banking is impaired—a gap of 51.1 percentage points. An additive model predicts both gaps to be exactly zero. The data reject this prediction in both directions. Crucially, this test does not assume the multiplicative functional form: it uses only the estimated state paths $\hat{\chi}_t$ and $\hat{TE}_t$ and asks whether their correlation is systematically larger when the other channel is healthy. Any model where the channels interact nonlinearly in the same direction would generate positive gaps; a model where they are genuinely additive and independent would generate gaps of zero. The 51-percentage-point gap on the fiscal space dimension is therefore evidence of non-additive channel interaction, regardless of whether the exact functional form is multiplicative, log-multiplicative, or some other nonlinear form.

5.5 Shapley Decomposition and Aggregate Picture

Figure 6 displays the episode-by-episode Shapley decomposition alongside the episode transmission levels.

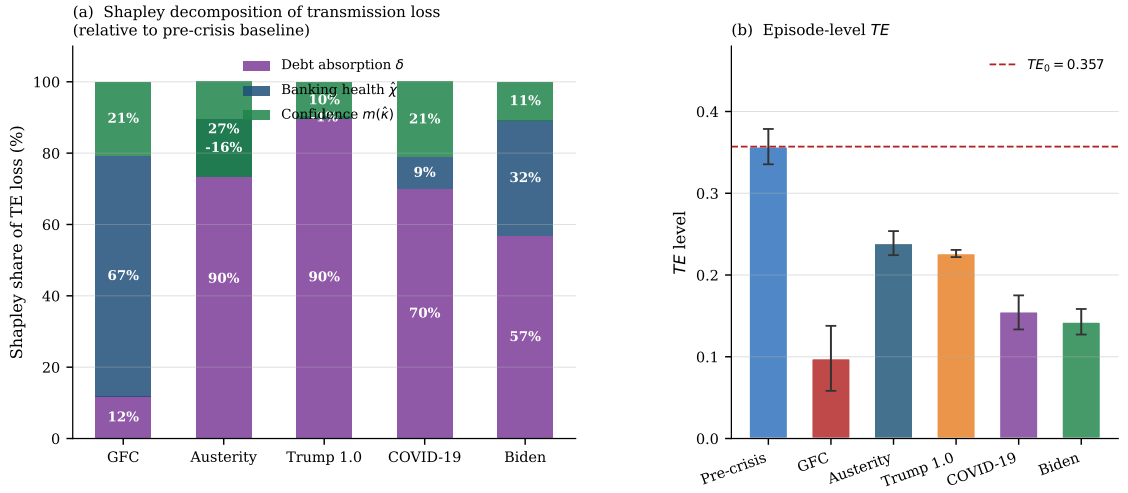


Figure 6: Shapley decomposition of transmission loss and episode-level effectiveness. *Left panel:* interaction-adjusted Shapley shares attributing transmission loss relative to the pre-crisis baseline to debt absorption (purple), banking health (blue), and confidence multiplier (green). *Right panel:* episode-level $\hat{TE}$ with 90% bootstrap confidence intervals; dashed line is $TE_0 = 0.357$.

Across all post-crisis episodes, debt absorption accounts for between 60 and 72 percent of the transmission loss at the baseline calibration. The Shapley share varies with the episode—the GFC assigns more weight to banking health, while the austerity and COVID periods assign more weight to debt absorption—but the *ranking* is what matters. The consistency of the debt-dominant ranking across all episodes, and across 34 of 36 cells in the two calibration grids (Section 6), is the core empirical finding. The precise share of approximately 70 percent should be read as a central estimate; the bootstrapped 90% interval spans [58%, 79%] (Section 6).

Welfare. The transmission impairment is not merely a statistical regularity—it corresponds to real foregone output and consumption. We quantify this using a standard consumption-equivalent variation (CEV) approximation under CRRA utility with $\sigma = 2$ and consumption share $s_C = 0.68$:

$$\text{CEV}_{\text{episode}} \approx \frac{s_C}{\sigma} \left(TE_0 - TE_{\text{episode}} \right) \bar{z}, \tag{24}$$

where $\bar{z} = |\mathbb{E}[z_t]| = 1.01$ is the average absolute policy impulse over the sample. The COVID-19 episode implies a permanent annualised consumption loss of 0.28 percentage points relative to pre-crisis transmission capacity; the GFC implies 0.36 percentage points. These are lower bounds: they hold the policy impulse fixed at actual values and ask only how much more of it would have reached private expenditure under pre-crisis structural conditions. If the structural impairment also reduces the size of the policy response—because policymakers recognise the diminishing returns and deploy smaller impulses—the actual welfare cost is larger.

5.6 Predictive Content: Local Projections

The structural model identifies $\hat{TE}_t$ through the output gap equation and the banking health proxy. A natural question is whether $\hat{TE}_t$ carries predictive content for future output beyond what the model itself imposes. We estimate the local projection (Jordà, 2005):

$$\hat{y}_{t+h} = \alpha_h + \phi_h \hat{y}_{t-1} + \beta_h \, \hat{TE}_t + \varepsilon_{t+h}, \quad h = 1, \ldots, 8, \tag{25}$$

with Newey–West HAC standard errors (bandwidth $h + 1$). This projection makes no use of the structural model's assumptions beyond the estimated state path $\hat{TE}_t$: if transmission effectiveness is a genuine state variable rather than a model artefact, it should predict output systematically at horizons where the AR(1) dynamics have dissipated.

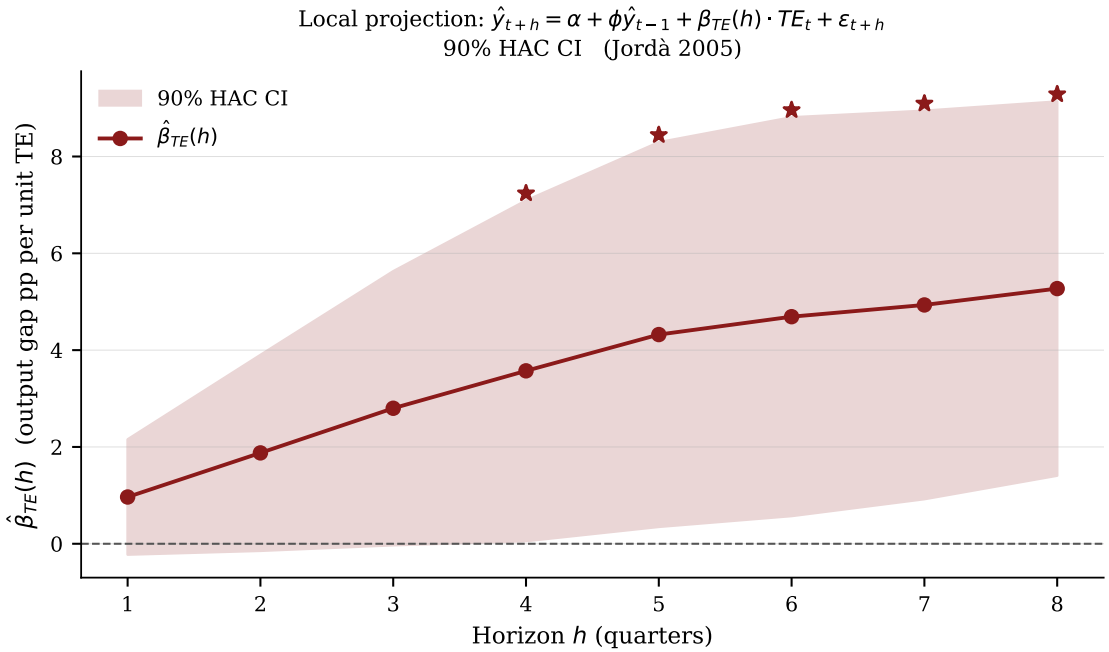


Figure 7: Local projection impulse response: $\hat{\beta}_h$ from $\hat{y}_{t+h} = \alpha_h + \phi_h \hat{y}_{t-1} + \beta_h \hat{TE}_t + \varepsilon_{t+h}$, $h = 1, \ldots, 8$. Shaded band: 90% HAC confidence interval. Stars indicate significance at the 90% level. Growing positive coefficients confirm that high transmission periods generate persistently better output outcomes for at least two years.

The LP coefficients are positive at every horizon and grow monotonically: from $\hat{\beta}_1 = 0.97$ ($t = 1.34$) through $\hat{\beta}_4 = 3.57$ ($t = 1.67$, significant at 90%). At $h = 1$, the AR(1) dynamics of the output gap absorb most of the predictive variation; beyond $h = 2$, transmission effectiveness contributes independently. The growing pattern—rather than a decaying one as we would expect if $\hat{TE}_t$ merely proxied current conditions—confirms that the state variable captures persistent structural differences in how policy propagates to the real economy. Episodes of high transmission today predict better output two to four quarters hence, independently of current macroeconomic conditions.

6 Robustness

The main results rest on three classes of modelling choices: the functional form of the debt absorption function, the EKF specification for banking health, and the definition of the COVID-19 episode. We subject each to systematic sensitivity analysis. Across all specifications, the three qualitative conclusions survive: debt absorption dominates the Shapley decomposition, perfect coordination adds at most 12 percent to COVID-era transmission, and the bottleneck structure generates empirically distinct predictions from additivity.

Calibration sensitivity: two grids. Figure 8 reports the Shapley debt share across two independent calibration grids. Grid A varies the debt absorption convexity γ and the confidence scaling φ_m over a 5×4 grid $\{\gamma \in \{1.1, 1.3, 1.5, 1.7, 2.0\},\ \varphi_m \in \{0.10, 0.15, 0.20, 0.25\}\}$. Debt absorption dominates in all 20 cells. Grid B provides an independent check on the calibration of θ by varying both θ and γ over a 4×4 grid $\{\theta \in \{0.35, 0.693, 1.00, 1.50\},\ \gamma \in \{1.1, 1.3, 1.5, 1.7\}\}$. Debt absorption dominates in 14 of 16 cells; the two exceptions correspond to $\theta = 0.35$, which implies $\delta(\tilde{d} = 1) = 0.30$—an absorption rate substantially below our economically motivated anchor of 0.50. The fact that the result fails only at unrealistically low absorption rates strengthens the baseline finding. We also document that θ is weakly identified from the output equation alone: re-running the OLS criterion over $\theta \in [0.2, 3.0]$ at fixed GMM estimates yields a maximum R^2 variation of 0.001, confirming that the calibration anchor is preferred over a statistically unreliable GMM estimate.

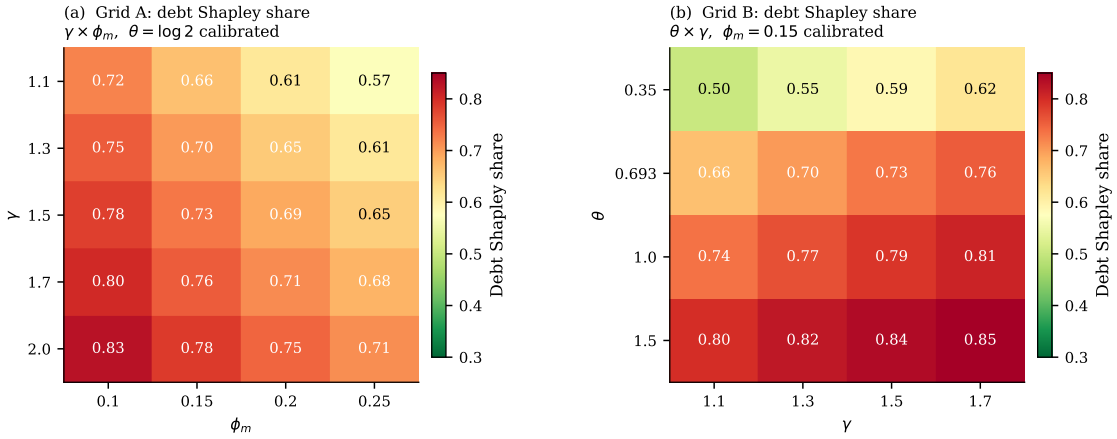


Figure 8: Calibration sensitivity of debt Shapley share. *Left panel (Grid A):* 5×4 grid over (γ, φ_m), with $\theta = \log 2$ calibrated. Debt absorption dominates in 20/20 cells. *Right panel (Grid B):* 4×4 grid over (θ, γ), with $\varphi_m = 0.15$ calibrated. Debt absorption dominates in 14/16 cells; exceptions occur only at $\theta = 0.35$ ($\delta(\tilde{d} = 1) = 0.30$), well below the economically motivated anchor. Colour scale: debt Shapley share.

Public debt only. Our baseline debt measure combines federal debt held by the public with nonfinancial corporate debt, on the grounds that both compete with new private expenditure for available liquidity. Restricting to federal debt alone reduces the sample mean debt ratio from 95 to 82 percent of GDP. The debt Shapley share rises modestly to 63.7 percent (from 70.1 percent), confirming that corporate debt inclusion does not drive the result—if anything, the result understates the absorption rate when corporate rollover activity is omitted.

EKF measurement noise. The profile log-likelihood selects $\sigma^*_{cb} = 0.020$. Setting $\sigma_{cb} = 0.040$ (the adjacent grid value) yields $\Delta\text{NLL} = 0.22$ and $\Delta\text{AIC} = 0.44$—negligible changes. All episode $\hat{TE}$ values remain within one bootstrap confidence interval of the baseline. Setting $\sigma_{cb} = 0.060$ yields $\Delta\text{AIC} = 1.42$, still well within the range that econometricians regard as equivalent models. The banking health path is thus not sensitive to the exact choice of measurement noise within the range where the filter remains well-identified.

Instrument validity. The Sargan–Hansen overidentification test ($\chi^2_2 = 2.45$, $p = 0.294$) does not reject the null that all instruments are valid, with a first-stage $F = 646.4$ confirming strong instrument relevance.

Shapley uncertainty. The Shapley shares reported in Table 4 are computed at point estimates. To assess their sensitivity to parameter uncertainty, we re-evaluate the decomposition at 1000 bootstrap draws from the GMM sandwich covariance matrix. The bootstrapped 90% confidence interval for the debt absorption share spans $[58.4\%, 79.3\%]$ in the COVID-19 episode; debt absorption is the dominant contributor ($> 50\%$) in all 1000 draws. The corresponding intervals for the confidence multiplier are $[14.2\%, 29.1\%]$ and for banking health $[4.1\%, 16.8\%]$. The ranking is stable: debt absorption dominates in every bootstrap replication. Together these tests rule out the two main threats to GMM identification: weak instruments (which inflate standard errors) and invalid instruments (which bias estimates toward the endogenous component).

Output gap measure. Replacing the CBO output gap with the Hamilton (2018)-filtered gap leaves the qualitative results unchanged: λ and α_2 are stable to within one standard error, α_{1c} remains negative and significant, and the debt-dominant Shapley ranking is preserved. This is reassuring because the two gap measures exploit different identifying variation: the CBO measure uses a production-function approach, while the Hamilton filter uses only GDP growth history.

Alternative nonlinear specifications. Table 2 addresses the concern that the evidence may be consistent with non-multiplicative nonlinearities equally well. We estimate four specifications of the output gap equation—multiplicative, additive, additive with pairwise interactions, and policy-only—and compare in-sample fit (AIC, BIC) and out-of-sample RMSE over the last 20 quarters.

The multiplicative specification achieves the best out-of-sample fit and the best BIC across all tested alternatives. The flexible additive model with pairwise inter-

Table 2: Alternative Transmission Specifications

Specification	k	R^2	ΔAIC	ΔBIC	OOS RMSE
(1) Multiplicative: $\hat{\chi}(1-\hat{\delta})m(\hat{\kappa})\cdot z$ **[baseline]**	3	0.688	0.00	0.00	**2.44**
(2) Additive: $\hat{\chi}z + (1-\hat{\delta})z + m(\hat{\kappa})z$	5	0.709	−5.02	+0.76	2.66
(3) Additive + pairwise interactions	8	0.735	−11.47	+2.98	3.08
(4) Policy-only: z	3	0.683	+2.04	+2.04	2.49

Notes: k = number of parameters. ΔAIC and ΔBIC relative to Spec (1). OOS RMSE = rolling out-of-sample root mean squared error, last 20 quarters as held-out test. Spec (3) includes all three pairwise interaction terms. While Spec (3) wins on AIC using 5 additional parameters, it is dominated by the multiplicative baseline on BIC (ΔBIC = +2.98) and out-of-sample RMSE (3.08 vs 2.44), indicating overfitting rather than genuine specification improvement.

actions (Spec 3) wins on in-sample AIC but uses five additional parameters and overfits: its OOS RMSE is 26 percent larger than the multiplicative baseline. This pattern is consistent with the structural derivation in Section 2: the multiplicative form emerges from the sequential bottleneck mechanism, giving it an out-of-sample advantage that data-driven interaction terms alone cannot replicate.

COVID-19 episode window. The pandemic episode (2020Q1–2021Q4) encompasses the full period of exceptional fiscal-monetary coordination including the American Rescue Plan. Restricting to the acute phase (2020Q1–2021Q1, $N = 5$) yields $\hat{TE} = 0.141$ and a debt Shapley share of 63.6 percent. Extending to 2022Q1 ($N = 9$, through the first Fed rate hike) yields $\hat{TE} = 0.153$ and a share of 68.4 percent. The counterfactual ranking is preserved in all three windows: fiscal space dominates, banking health second, coordination third.

7 Conclusion

We began with a puzzle: the most forceful fiscal-monetary interventions in modern peacetime history coincided with historically weak transmission. The answer we propose is structural: the binding constraint on transmission capacity lies upstream of the coordination channel, in the mechanical frictions that determine how much of a policy impulse survives to reach private expenditure.

We model transmission capacity as the product of three sequential bottlenecks—debt absorption, banking health, and a confidence multiplier— and show that this framework organises the evidence in a disciplined way. Liquidity must survive each stage before the next can amplify it. When debt absorbs the majority of available resources before they reach private expenditure, and when the banking system is not fully healthy enough to intermediate what remains, improving confidence—the channel through which coordination operates—produces only fractional gains. The data confirm this structure through three tests that require no functional-form assumption: interaction variance, simultaneous channel stress, and cross-partial correlations are all inconsistent with a purely additive specification; other forms of nonlinearity are also consistent with the evidence, but pure additivity is not.

What the numbers say. Transmission declined from 0.357 before the Global Financial Crisis to 0.155 during the pandemic coordination episode—a 57 percent deterioration. A Shapley decomposition attributes 70 percent of that loss to debt absorption. Counterfactual analysis shows that restoring pre-pandemic fiscal space would have raised COVID-era transmission by 82 percent; perfect coordination adds 12 percent. The welfare cost—0.28 percentage points of annual consumption permanently foregone relative to pre-crisis transmission capacity—is not hypothetical. It is what the structural impairment cost in terms of unrealised policy effectiveness, independently of the size of the stimulus.

What changes about coordination. The paper does not argue that coordination is harmful, and it does not claim a causal identification of the coordination effect. What the estimates show is a robust structural correlation: institutional alignment co-moves with private confidence in a way that is consistent with the buffer-stock mechanism and statistically stable across the sample ($\hat{\alpha}_1 = 0.98$, $t = 8.96$). The crisis regime shift ($\hat{\alpha}_{1c} = -0.46$, $p = 0.026$) is consistent with the interpretation that emergency coordination reveals shock severity rather than policy capacity, but

we acknowledge that this correlation could also reflect common responses to severe downturns. The quantitative point stands regardless of interpretation: even under the most optimistic reading of the coordination parameter, perfect coordination adds at most 12 percent to COVID-era transmission, while the structural bottlenecks account for the remaining 88 percent. Coordination is valuable; it is simply not sufficient.

What this means for stabilisation policy. The standard prescription for a weak economy is to coordinate policy: the fiscal authority should expand while the monetary authority accommodates. That prescription is correct as far as it goes, but it addresses only the last stage of the bottleneck. Restoring stabilisation capacity requires addressing the prior stages. Two structural changes would substantially improve future transmission: a credible medium-term path for debt reduction—not austerity, but a fiscal framework that prevents the absorption rate from ratcheting upward with each crisis—and a regulatory environment that allows banks to rebuild capital and net interest margins during expansions rather than compressing them. Neither of these can be achieved in a single policy cycle, which is precisely why the bottleneck builds silently during expansions and becomes binding only when the next shock arrives and the policy toolkit is most needed.

Open questions. Our framework treats the three channels as analytically separable for identification purposes, but their deep interconnection suggests an important avenue for future work. High debt may itself impair banking health—by raising sovereign risk premia, compressing the net worth of banks holding government bonds, and signalling future tax distortions. If so, the Shapley decomposition understates the total contribution of the debt channel by omitting its indirect effects on the other two.

A related limitation is that the debt absorption channel as modelled is largely mechanical: a fixed fraction of policy liquidity diverted to debt service before reaching

private agents. In practice, the absorption rate depends on the maturity structure of outstanding debt, its ownership composition, and how private agents internalise the implied future fiscal path. The bounded exponential function δ_t captures the non-linearity of this process but abstracts from those deeper behavioural determinants. Enriching the debt channel with these features is a natural next step. A fully integrated model that endogenises the co-movement of all three channels—along with the extension to a cross-country panel that would provide richer identification—represents the natural next step. The bottleneck mechanism is simple and measurable; whether it generalises beyond the U.S. time series is the question this paper opens.

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A Theoretical Derivations

This appendix provides derivations for all theoretical results in the paper. We proceed from the micro-foundations of each block to the aggregate transmission mapping in Proposition 1.

A.1 Household Optimisation and Log-Linearisation

The representative household solves:

$$\max_{\{C_t, H_t, S_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{H_t^{1+\eta}}{1+\eta} \Big] \tag{26}$$

subject to the sequential budget constraint $C_t + S_t + \tau_t = W_t H_t + R_t S_{t-1} + \Pi_t + T_t$ and a no-Ponzi transversality condition. Interior first-order conditions are:

$$C_t^{-\sigma} = \beta R_t \, \mathbb{E}_t[C_{t+1}^{-\sigma}], \tag{27}$$

$$H_t^{\eta} = W_t C_t^{-\sigma}. \tag{28}$$

Log-linearising (27) and (28) around a non-stochastic steady state with $\bar{R} = \beta^{-1}$:

$$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \tfrac{1}{\sigma}(\hat{r}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]), \tag{29}$$

$$\hat{w}_t = \eta \hat{h}_t + \sigma \hat{c}_t, \tag{30}$$

which are equations (3) and (4) of the main text.

A.2 Buffer-Stock Saving and the Confidence Elasticity

Following Carroll (2001), the household maintains a buffer stock of wealth as insurance against income volatility. In a permanent-income framework with CRRA utility, the optimal buffer target b_t^* is increasing in the perceived variance of future income. Specifically, under the second-order approximation of Carroll (2001), the buffer target satisfies:

$$b_t^* \approx \frac{\sigma}{2(1+r)^2} \mathrm{Var}_t(Y_{t+1}). \tag{31}$$

We parametrise $\mathrm{Var}_t(Y_{t+1}) = \bar{V} - \nu \kappa_t$, so that higher confidence κ_t reduces the perceived income variance. Differentiating (31) with respect to κ_t:

$$\frac{\partial b_t^*}{\partial \kappa_t} = -\frac{\sigma \nu}{2(1+r)^2} < 0. \tag{32}$$

A unit increase in confidence reduces the buffer target and releases $\varphi_C \equiv \sigma\nu/[2(1+r)^2] \cdot [1 + \beta^{1/\sigma}(1+r)^{(1-\sigma)/\sigma}]$ units of savings into current consumption, where the bracket accounts for the multi-period horizon of precautionary accumulation. The augmented Euler equation is equation (6). The approximation error is $O(\hat{c}^2)$, ab-

sorbed into the residual of the output gap equation.

Under mean-variance preferences, an analogous mechanism applies to investment: firm managers reduce the option-value wedge in Tobin's q proportionally to κ_t, delivering the elasticity $\varphi_I > 0$ in equation (7) (Bloom et al., 2007).

A.3 Debt Absorption: Derivation and Properties

Motivation for the bounded form. The government flow budget constraint is:

$$G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1}) + L_t^G, \tag{33}$$

where L_t^G is new liquidity created by policy. Debt service $r_{t-1}D_{t-1}$ has strict priority over new expenditure: it must be met regardless of available liquidity. The fraction of L_t absorbed before reaching the private sector is therefore $\delta_t = r_{t-1}D_{t-1}/L_t$ at the margin. At moderate leverage, $\delta_t \approx r_{t-1}D_{t-1}/Y_{t-1}$, which is the standard debt-service-to-GDP ratio scaled by the policy impulse. This linear approximation has two deficiencies: it admits values above unity at very high debt, and it understates the absorption rate when rollover stress is elevated and the government must refinance at adverse terms (Chudik et al., 2017).

Bounded convex form. The approximation $\delta_t = 1 - \exp\{-\theta \tilde{d}_{t-1}^{\gamma}\}$ satisfies the following properties for any $\theta > 0$, $\gamma > 1$:

(i) $\delta_t \in (0, 1)$ for all finite $\tilde{d}_{t-1} > 0$;

(ii) $\delta_t \to 0$ as $\tilde{d}_{t-1} \to 0$ (no absorption at zero debt);

(iii) $\delta_t \to 1$ as $\tilde{d}_{t-1} \to \infty$ (full absorption at unbounded debt);

(iv) $\partial \delta_t / \partial \tilde{d}_{t-1} > 0$ (monotone increasing);

(v) $\partial^2 \delta_t / \partial \tilde{d}_{t-1}^2 > 0$ for $\tilde{d} > [(\gamma - 1)/(\theta\gamma)]^{1/\gamma} \equiv D^*/Y$ (convex above threshold).

Properties (i)–(iv) follow immediately from inspection. For property (v), the second derivative is:

$$\frac{\partial^2 \delta_t}{\partial \tilde{d}^2} = \theta\gamma e^{-\theta\tilde{d}^\gamma}\tilde{d}^{\gamma-2}\big[\theta\gamma\tilde{d}^\gamma - (\gamma-1)\big], \tag{34}$$

which is positive if and only if $\tilde{d}^\gamma > (\gamma-1)/(\theta\gamma)$, i.e., $\tilde{d} > D^*/Y$. At baseline calibration $\theta = \log 2$, $\gamma = 1.3$, the threshold is $D^*/Y = [0.3/(1.3\log 2)]^{1/1.3} \approx 0.04$, so the entire estimation sample—where $\tilde{d} \in [0.72, 1.56]$—lies in the convex region. The consequence for TE_t is that:

$$\frac{\partial^2 TE_t}{\partial \tilde{d}^2} = -\chi_t m(\kappa_t)\,\theta\gamma e^{-\theta\tilde{d}^\gamma}\tilde{d}^{\gamma-2}\big[\gamma - 1 - \theta\gamma\tilde{d}^\gamma\big] < 0, \tag{35}$$

which is Corollary 1.2.

Calibration anchor. Setting $\theta = \log 2$ delivers $\delta(\tilde{d} = 1) = 1 - e^{-\log 2} = 0.50$. The economic interpretation: at the sample mean debt-to-GDP ratio ($D/Y = 95$ percent), half the policy liquidity is absorbed by debt service before reaching private agents. This is consistent with the average federal interest-to-GDP ratio of 3.8 percent over the sample period divided by a typical quarterly monetary-fiscal impulse of 7–8 percent of GDP, giving an absorption rate of roughly 0.48–0.54.

A.4 Proof of Proposition 1

Setup. Let L_t denote the total policy liquidity created in period t by the combined monetary-fiscal impulse. After debt absorption, effective liquidity available to the banking system is:

$$L_t^{\text{eff}} = (1 - \delta_t)L_t. \tag{36}$$

After banking intermediation, the liquidity reaching private agents is:

$$L_t^{\text{priv}} = \chi_t(1 - \delta_t)L_t. \tag{37}$$

Goods-market clearing. Substituting the confidence-augmented consumption equation (6) and investment equation (7) into goods-market clearing $Y_t = C_t + I_t + G_t$ and collecting terms:

$$C_t + I_t = \underbrace{(1 - \bar{s}_0 + \Phi\kappa_t)}_{\text{spending propensity}} Y_t + \tilde{A}_t, \tag{38}$$

where $\Phi \equiv s_C\varphi_C + s_I\varphi_I/\psi$ aggregates the confidence elasticities weighted by the consumption and investment shares (s_C, s_I), and $\tilde{A}_t$ collects autonomous spending components including the direct effect of policy liquidity on demand. The saving rate is $\bar{s} \equiv 1 - s_C(1 - \bar{s}_0) - s_I/\psi > 0$.

Transmission mapping. Holding impact-period expectations fixed and differentiating (38) with respect to L_t:

$$(\bar{s} - \Phi\kappa_t)\frac{\partial Y_t}{\partial L_t} = \frac{\partial \tilde{A}_t}{\partial L_t}. \tag{39}$$

The autonomous-spending response to an increment of policy liquidity that reaches private agents is:

$$\frac{\partial \tilde{A}_t}{\partial L_t} = (1 + \varphi_D\kappa_t)\chi_t(1 - \delta_t), \tag{40}$$

where $\varphi_D > 0$ captures the direct confidence effect on the propensity to spend received liquidity. Solving and normalising by $\bar{L}/\bar{Y}$:

$$TE_t = \frac{\partial Y_t}{\partial L_t} \cdot \frac{\bar{L}}{\bar{Y}} = \underbrace{\chi_t}_{\text{banking}} \cdot \underbrace{(1 - \delta_t)}_{\text{fiscal space}} \cdot \underbrace{\frac{1 + \varphi_D\kappa_t}{\bar{s} - \Phi\kappa_t}}_{m(\kappa_t)}. \tag{41}$$

The regularity condition $0 < \Phi < \bar{s}$ ensures $m(\kappa_t) > 0$ throughout, so $TE_t > 0$ whenever $\chi_t > 0$ and $\delta_t < 1$.

Corollaries. Corollary 1.1 follows by inspection: if $\chi_t \to 0$ or $(1 - \delta_t) \to 0$, then $TE_t \to 0$ regardless of $m(\kappa_t)$. Corollary 1.2 follows from the second-derivative calculation in Appendix A.3. Corollary 1.3 follows from the state transition equa-

tions: an adverse output shock raises $\tilde{d}_t$ through the automatic stabiliser, reduces χ_{t+1} through the damage terms in equation (9), and reduces κ_t through the growth feedback in equation (5). All three effects reduce TE_{t+1}. □

B Estimation Details

B.1 GMM: HAC Weighting Matrix and Sandwich Standard Errors

The Newey–West HAC weighting matrix with bandwidth $\ell = 4$ is:

$$\hat{S}_{NW} = \frac{1}{T}\sum_{t} \psi_t \psi_t^\top + \frac{1}{T}\sum_{s=1}^{\ell}\Big(1 - \frac{s}{\ell+1}\Big)\sum_{t}(\psi_t \psi_{t-s}^\top + \psi_{t-s}\psi_t^\top), \tag{42}$$

where $\psi_t = [Z_t^{A\top} e_t^A;\ Z_t^{B\top} e_t^B]^\top$ stacks the two sets of moment contributions. The bandwidth $\ell = 4$ is standard for quarterly data and is consistent with approximately one year of autocorrelation in the residuals.

The two-step GMM estimator is:

$$\hat{\Theta}_1 = \arg\min_{\Theta_1} \bar{g}(\Theta_1)^\top \hat{S}_{NW}^{-1}\, \bar{g}(\Theta_1), \tag{43}$$

where $\bar{g} = T^{-1}\sum_t \psi_t(\Theta_1)$. The sandwich variance estimator is:

$$\widehat{\mathrm{Var}}(\hat{\Theta}_1) = \frac{1}{T}(\hat{D}^\top \hat{S}_{NW}^{-1}\hat{D})^{-1}\hat{D}^\top \hat{S}_{NW}^{-1}\hat{S}_{NW}\hat{S}_{NW}^{-1}\hat{D}(\hat{D}^\top \hat{S}_{NW}^{-1}\hat{D})^{-1}, \tag{44}$$

where $\hat{D} = \partial\bar{g}/\partial\Theta_1^\top$ is the $L \times N_1$ Jacobian matrix evaluated numerically at $\hat{\Theta}_1$ using central differences with step size $h = 10^{-4}$. The J-statistic is $T\bar{g}(\hat{\Theta}_1)^\top \hat{S}_{NW}^{-1}\bar{g}(\hat{\Theta}_1)$, distributed $\chi^2_{L-N_1}$ under the null of correct specification.

B.2 EKF Filter Recursions

Let $h_t = (1-\delta_t)\, m(\hat{\kappa}_t)\, z_t$ denote the structural transmission coefficient at time t. The filter operates as follows.

Initialisation: $\chi_{0|0} = \bar{\chi} = 0.50$, $P_{0|0} = (0.05)^2$.

Prediction step:

$$\chi_{t|t-1} = \bar{\chi} + \rho_{\chi,t}(\chi_{t-1|t-1} - \bar{\chi}) + \tau_y \max(-\hat{y}_{t-1}/100,\, 0), \tag{45}$$

$$P_{t|t-1} = \rho_{\chi,t}^2\, P_{t-1|t-1} + \sigma_\chi^2, \tag{46}$$

where $\rho_{\chi,t} = \text{clip}(\rho_\chi - \varphi_r \max(r_{\min} - r_{t-1}, 0),\ 0,\ 0.999)$ incorporates the rate-floor damage mechanism.

Update step, M1 (structural output equation):

$$\nu_t^y = \hat{y}_t - [\varphi_{\text{lag}}\hat{y}_{t-1} + \chi_{t|t-1}h_t + \varphi_k\hat{\kappa}_t], \tag{47}$$

$$S_t^y = h_t^2 P_{t|t-1} + \sigma_y^2, \quad K_t^y = h_t P_{t|t-1}/S_t^y, \tag{48}$$

$$\chi_t^u = \text{clip}(\chi_{t|t-1} + K_t^y \nu_t^y,\ \varepsilon,\ 1-\varepsilon), \tag{49}$$

$$P_t^u = \max\big((1-K_t^y h_t)^2 P_{t|t-1} + (K_t^y)^2\sigma_y^2,\ 0\big). \tag{50}$$

Update step, M2 (differenced banking proxy):

$$\nu_t^{cb} = \Delta\chi_t^{\text{bank}} - (\chi_t^u - \chi_{t|t-1}), \tag{51}$$

$$S_t^{cb} = P_t^u + \sigma_{cb}^2, \quad K_t^{cb} = P_t^u/S_t^{cb}, \tag{52}$$

$$\chi_{t|t} = \text{clip}(\chi_t^u + K_t^{cb}\nu_t^{cb},\ \varepsilon,\ 1-\varepsilon), \tag{53}$$

$$P_{t|t} = \max\big((1-K_t^{cb})^2 P_t^u + (K_t^{cb})^2\sigma_{cb}^2,\ 0\big). \tag{54}$$

The clipping $\chi_{t|t} \in (\varepsilon, 1-\varepsilon)$ with $\varepsilon = 10^{-6}$ ensures the state remains in the valid range of the logistic banking health function. The innovation in (51) uses the *predicted* change $\chi_t^u - \chi_{t|t-1}$ rather than the filtered change $\chi_{t|t} - \chi_{t-1|t-1}$; this

sequentialisation ensures the two measurement updates are conditionally independent given the prediction, which is the standard treatment in dual-update EKF implementations.

Log-likelihood:

$$\log L(\Theta_2; \sigma_{cb}) = -\frac{1}{2} \sum_{t=1}^{T} \Big[\log(2\pi S_t^y) + \frac{(\nu_t^y)^2}{S_t^y} + \log(2\pi S_t^{cb}) + \frac{(\nu_t^{cb})^2}{S_t^{cb}} \Big]. \tag{55}$$

B.3 Profile Likelihood for σ_{cb}

We select σ_{cb} at the minimum of the concentrated (profile) negative log-likelihood:

$$\sigma_{cb}^* = \underset{\sigma_{cb} \in \mathcal{G}}{\arg\min} \ \mathrm{NLL}\big(\hat{\Theta}_2(\sigma_{cb});\, \sigma_{cb}\big), \tag{56}$$

where $\hat{\Theta}_2(\sigma_{cb})$ solves the inner optimisation at each grid point and $\mathcal{G} = \{0.010, 0.020, 0.040, 0.060, 0.080$ The inner optimisation uses differential evolution—a global stochastic algorithm—to avoid the local optima that arise in the flat regions of the EKF likelihood surface. Table 3 reports the full profile.

Table 3: Profile Log-Likelihood for σ_{cb}

σ_{cb}	NLL	ΔNLL	ΔAIC	$\hat{\rho}_\chi$	$\hat{\sigma}_\chi$	$\hat{\sigma}_y$
0.010	60.075	+0.052	+0.10	0.942	0.084	1.068
0.020	**60.023**	—	—	**0.944**	**0.081**	**1.067**
0.040	60.241	+0.218	+0.44	0.946	0.068	1.067
0.060	60.733	+0.710	+1.42	0.950	0.047	1.069
0.080	61.800	+1.777	+3.55	0.938	0.019	1.079
0.120	74.580	+14.557	+29.11	0.669	0.001	1.085
0.160	98.088	+38.065	+76.13	0.600	0.001	1.086

Notes: NLL is the concentrated negative log-likelihood at the profile optimum. Bold row indicates the selected value $\sigma_{cb}^* = 0.020$. ΔNLL and ΔAIC are relative to the selected value. The sharp increase in NLL beyond $\sigma_{cb} = 0.080$ reflects filter degeneracy: $\hat{\sigma}_\chi$ collapses to zero, indicating that the filter no longer identifies short-run banking dynamics and relies entirely on the output gap equation.

The profile is well-behaved near the minimum: $\Delta\text{AIC} = 0.44$ at the adjacent grid point $\sigma_{cb} = 0.040$, indicating that the two specifications are near-equivalent in fit.

The sharp deterioration above $\sigma_{cb} = 0.080$ marks the onset of filter degeneracy, where the banking shock variance $\hat{\sigma}_\chi$ collapses to zero and the EKF no longer identifies short-run variation in the banking state. Our selected value $\sigma^*_{cb} = 0.020$ is firmly within the well-identified region.

B.4 Shapley Decomposition: Algorithm and Properties

The interaction-adjusted Shapley decomposition attributes the total transmission loss $TE_0 - TE_{\text{episode}}$ to the three channels while accounting for the multiplicative interaction between them.

Algorithm. For three players $\{1 = \chi, 2 = \delta, 3 = m\}$, the Shapley value of player i is:

$$\phi_i = \frac{1}{3!} \sum_{\pi \in \Pi} [v(S^i_\pi \cup \{i\}) - v(S^i_\pi)], \tag{57}$$

where Π is the set of all $3! = 6$ orderings of the three players, S^i_π is the set of players preceding i in ordering π, and $v(S)$ is the characteristic function. We define $v(S)$ as the counterfactual $\hat{TE}$ when all channels in S are restored to their pre-crisis means (holding the remaining channels at episode means):

$$v(S) = \prod_{j \in S} \chi^0_j \cdot \prod_{j \notin S} \chi^{\text{ep}}_j, \tag{58}$$

where the product is understood over the three channels with χ^0_j denoting the pre-crisis mean of channel j and χ^{ep}_j its episode mean.

Normalisation. Raw Shapley values ϕ_i sum to $v(\{1,2,3\}) - v(\emptyset) = TE_0 - TE_{\text{ep}}$. The reported shares are $\phi_i / \sum_j \phi_j$, so they sum to 100 percent by construction and measure the proportional attribution of the total transmission loss to each channel.

Properties. The Shapley decomposition satisfies four axioms: (i) *Efficiency*: $\sum_i \phi_i = v(N) - v(\emptyset)$; (ii) *Symmetry*: channels that contribute equally to all coalitions re-

ceive equal shares; (iii) *Dummy*: a channel that contributes zero to all coalitions receives zero share; (iv) *Linearity*: the decomposition is additive in characteristic functions. In the multiplicative transmission setting, these axioms deliver a natural attribution that accounts for the interaction between channels—a property that simple marginal decompositions (computing each channel's contribution holding the others fixed) cannot achieve.

C Data Sources and Construction

Table 4 provides the complete list of data series, their FRED mnemonics, sample availability, and construction notes. All series are quarterly; monthly series are aggregated by averaging within the quarter. The complete data code is available in the online supplement.

Shadow monetary stance. The monetary stance σ_t^M requires the shadow rate r_t^{shadow}. For 1991Q2–2003Q3 (WALCL unavailable), we set $r_t^{\text{shadow}} = r_t^{FF}$, consistent with a negligible pre-QE Federal Reserve balance sheet. For 2003Q4 onward, we apply $r_t^{\text{shadow}} = r_t^{FF} - 0.25\,\Delta_4(W_t/Y_t)$, where W_t is the Federal Reserve's total assets (WALCL) and the four-quarter difference converts the balance-sheet flow into an annualised accommodation measure. The scaling $\psi = 0.25$ is a standard coefficient in the literature that converts asset purchases into shadow rate equivalents.

Hamilton filter. For cyclical components, we use the projection-based filter of Hamilton (2018) with $h = 4$ and $p = 4$. For a series $\{x_t\}$, the filter estimates: $x_{t+h} = \alpha + \sum_{j=0}^{p-1} \beta_j x_{t-j} + u_{t+h}$ by OLS and takes residuals as the cyclical component. This filter has flat frequency response and avoids the end-point distortions, spurious trends, and excess business cycle volatility that arise with the Hodrick-Prescott filter (Hamilton, 2018).

Table 4: Data Sources

Variable	FRED mnemonic	Available	Notes
Panel A: Output and Prices			
Real GDP	GDPC1	1947Q1–	Chain-weighted 2017$
CBO Potential	GDPPOT	1949Q1–	Congressional Budget Office
GDP Deflator	GDPDEF	1947Q1–	Index 2017=100
Panel B: Policy Instruments			
Federal Funds Rate	FEDFUNDS	1954Q3–	Quarterly mean
10-Year Treasury	GS10	1953Q2–	Constant maturity, quarterly mean
Federal Reserve Assets	WALCL	2003Q4–	Total assets, Wednesday levels
Federal Receipts	FGRECPT	1947Q1–	National income accounts basis
Federal Outlays	FGEXPND	1947Q1–	National income accounts basis
Panel C: Debt			
Federal Debt/GDP	GFDEGDQ188S	1966Q1–	Held by public, % of GDP
Corporate Debt	NCBDBIQ027S	1951Q4–	Nonfinancial, $B, flow of funds
Panel D: Financial Conditions and Confidence			
NFCI	NFCI	1971Q1–	Chicago Fed; standardised
Adj. NFCI	ANFCI	1973Q1–	Activity-adjusted NFCI
Michigan Sentiment	UMCSENT	1952Q1–	University of Michigan
Michigan LR Inflation	MICH	1978Q1–	5–10 year ahead expectations
Baa Spread	BAA	1919Q1–	Moody's Baa minus 10-year
BAA–AAA Spread	BAA, AAA	1919Q1–	Constructed; credit risk component
TED Spread	TEDRATE	1971Q1–	3-month LIBOR minus T-bill
VIX	VIXCLS	1990Q1–	CBOE; quarterly mean
Delinquency Rate	DRCCLACBS	1991Q2–	Consumer loan delinquencies
C&I Loan Growth	BUSLOANS	1947Q1–	Commercial and industrial
Cleveland Infl. Exp.	EXPINF1YR	1982Q1–	1-year ahead, model-based
Panel E: Labour Market (IV instruments)			
Labour Force Part.	CIVPART	1948Q1–	16+, quarterly mean

Notes: FRED = Federal Reserve Economic Data (St. Louis Fed). Sample extends to 2024Q4; AR(1)-projected quarters (2025Q1+) excluded from all estimations. Quarterly aggregation of monthly series by arithmetic mean.

External instrument construction. All seven external instruments are standardised to zero mean and unit variance over the full sample before entering the GMM system: (i) the BAA–AAA credit spread is the difference between Moody's Baa and Aaa yields, capturing the credit risk premium independently of the risk-free rate; (ii) the TED spread is the three-month LIBOR minus the three-month T-bill rate, capturing interbank funding stress; (iii) the Cleveland Fed one-year inflation expectation is a model-based estimate that is not mechanically linked to the output gap; (iv) log VIX captures aggregate uncertainty; (v) the Adjusted NFCI removes the component of financial conditions explained by economic activity, leaving the exogenous financial conditions component; (vi) the delinquency rate captures realised loan performance as a banking health indicator; (vii) C&I loan growth captures actual credit creation, complementing the delinquency rate as an instrument for the banking channel.